

Conformal Prediction: A Data Perspective

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Conformal prediction (CP), a distribution-free uncertainty quantification (UQ) framework, reliably provides valid predictive inference for black-box models. CP constructs prediction sets or intervals that contain the true output with a specified probability. However, modern data science's diverse modalities, along with increasing data and model complexity, challenge traditional CP methods. These developments have spurred novel approaches to address evolving scenarios. This survey reviews the foundational concepts of CP and recent advancements from a data-centric perspective, including applications to structured, unstructured, and dynamic data. We also discuss the challenges and opportunities CP faces in large-scale data and models.

CCS Concepts: • **Computing methodologies** → **Machine learning**; **Uncertainty quantification**;

Additional Key Words and Phrases: Natural language processing, Computer vision, Time series

ACM Reference Format:

Xiaofan Zhou, Baiting Chen, Yu Gui, and Lu Cheng. 2025. Conformal Prediction: A Data Perspective. *ACM Comput. Surv.* 58, 2, Article 49 (September 2025), 37 pages. <https://doi.org/10.1145/3736575>

1 Introduction

The recent rapid development of well-designed and powerful **machine learning (ML)** models has significantly transformed our lives. However, the success of these models is often evaluated based on the accuracy of their predictions, which, while important, is not sufficient in many real-world scenarios. In high-stakes applications, it is equally critical to assess the *uncertainty* of model outputs. **Uncertainty quantification (UQ)** has long been a central problem in fields like statistics and ML. Several well-established methods, such as Bayesian inference and resampling techniques, have been widely adopted to address UQ. However, Bayesian posterior intervals are only valid if the distributional assumptions of the model are correctly specified, which may not always be the case in practical applications. Similarly, when constructing prediction sets for unobserved outputs,

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This work is supported by the National Science Foundation (NSF) Grant #2312862, NSF-Simons SkAI Institute, National Institutes of Health (NIH) #R01AG091762, Amazon Research Award, and a Cisco gift grant.

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ACM 0360-0300/2025/09-ART49

<https://doi.org/10.1145/3736575>

resampling approaches based on the residuals of observed data can result in overly narrow intervals since the estimate of variance is too small [19], compromising the validity of the prediction set. This raises a fundamental question:

*Is it possible to provide a valid confidence or prediction set,
regardless of the model in use and with minimal assumptions about the data distribution?*

CP [186] is model-agnostic and can be applied to any ML model, generating prediction sets/intervals based on non-conformity scores derived from the model's outputs. CP offers several desirable features: (1) *Finite-sample validity*: CP generates calibrated prediction sets/intervals that ensure high-probability coverage with finite-sample theoretical guarantees, without simplifying the models' complexity or relying on strong assumptions about the data generating process. More concretely, the validity of CP only requires the *exchangeability* assumption of data. (2) *Lightweight*: As a wrapper method, CP does not require retraining the underlying model, making it computationally efficient and easy to apply to existing models. (3) *Versatility*: CP is model-agnostic, meaning it can be used with any model, regardless of its architecture or complexity. (4) *Informativeness*: When appropriately designed, CP's prediction set or interval can be sufficiently small to provide meaningful information about the original outputs.

1.1 CP from a Data Perspective

The increasing prevalence of big data, particularly with richer and more complex modalities, presents unique challenges for CP beyond traditional statistical settings. For instance: (1) **Generative models in natural language and image processing**: In these contexts, the target output can be ultra-high-dimensional or even infinite. For example, the output of a language model can be a permutation of any tokens in the pre-training corpus with arbitrary length, unlike the univariate real-valued responses in classical regression or classification tasks. For complex tasks like open-domain **question answering (QA)**, the output space is often unbounded [60, 170] (i.e., all possible text sequences). Adapting CP to handle these multi-dimensional and even infinite-dimensional outputs is essential to improve its applicability and predictive accuracy. (2) **Time series and streaming data**: These types of data are becoming increasingly important in fields such as economics, meteorology, geosciences, urban planning, and healthcare. However, time-dependent data poses significant challenges for CP, particularly because the assumption of data exchangeability is often violated in the presence of distribution shifts or temporal dependencies. (3) **Structured data** (e.g., groups, trees, and graphs): Structured data is typically quantitative and organized in predefined formats like rows and columns in relational databases while unstructured data (like text) is often qualitative and lacking a fixed schema [176, 210]. In ML applications involving structured data, canonical CP methods may fail to capture the intrinsic dependencies when the feature structure is ignored. Traditional inferential methods often rely on distributional assumptions specific to structured data. Thus, developing model-agnostic UQ tools that leverage implicit (or weighted) exchangeability within the data structure is crucial.

These challenges motivate this survey on CP *from a data perspective*, which introduces CP methods tailored to different data categories, as outlined in Figure 1. We categorize data into two main types based on the collection process: static data and dynamic data. Static data is further divided into structured data—including matrix/tensor data, hierarchical data, tree-structured data, and graph data—and unstructured data including image, text, and multi-modal data. Dynamic data is focused on spatio-temporal data, which is further subdivided into streaming, one-dimensional, and multi-dimensional categories. Note that dynamic data can be also structured, e.g., time-series data. This hierarchical organization provides a clear and systematic framework for understanding various data types, allowing for the appropriate application of CP techniques.

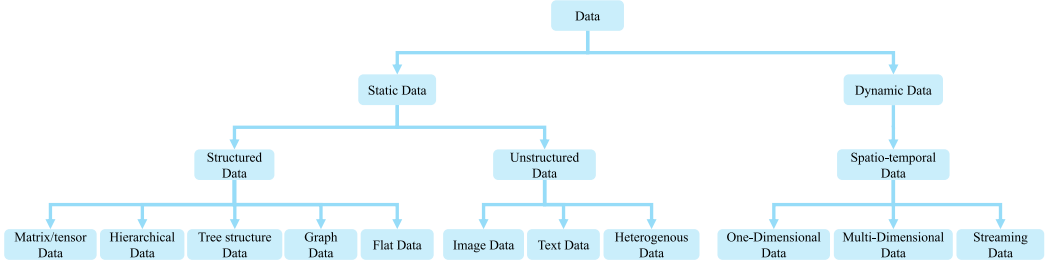


Fig. 1. The proposed taxonomy for CP from a data perspective.

1.2 Comparison with Existing Surveys

There are several existing surveys that either cover the theoretical foundations of CP with a review of more advanced algorithm developments (e.g., [9]) or review CP methods for a specific data type (e.g., [173]). Particularly, [9, 61] and [8] introduce theoretical foundations of CP including split CP, full CP, conformalized quantile regression, as well as recent advanced applications of CP. The survey paper [173] focuses on the application of CP in the specific field with spatio-temporal data and [26] centers around the field of natural language processing (NLP). [115] focuses CP’s application in autonomous systems, dealing with dynamic data by having trajectory calibration data. However, in light of the increasingly richer data types and modalities, a comprehensive and structured overview of CP from a data perspective is urgently needed to comprehensively understand the unique challenges of CP adapted to various data types. In this survey, we propose a novel taxonomy of CP from a data perspective (Figure 1) and review CP approaches and applications for structured data, unstructured data, and spatio-temporal data. We will start with a background introduction to basic CP and then provide a thorough discussion of CP across different tasks and data types, examining how the nature of the data influences the effectiveness of CP methods. Lastly, we summarize several open challenges and promising future directions that hope to spark future research in CP.

2 Background of CP

In this section, we first introduce basic CP approaches and important variants and then summarize common metrics in evaluating UQ methods. The major notations used throughout the survey are summarized in Table 2 in the supplementary.

2.1 Full CP

Let X and Y denote features and label, respectively. Taking the regression problem as an example, consider observed data $(X_1, Y_1), \dots, (X_n, Y_n)$ and the test data (X_{n+1}, Y_{n+1}) with Y_{n+1} unobserved. Our goal is to construct a prediction set for Y_{n+1} . Suppose that (X_i, Y_i) , $i = 1, \dots, n + 1$, are exchangeable (a relaxed assumption of i.i.d.¹), i.e., the joint distribution remains unchanged under any permutation, we have the property such that for any score function $V : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$,

$$\{V(X_{n+1}, Y_{n+1}) \mid \{V(X_i, Y_i)\}_{i=1}^{n+1} = \{v_i\}_{i=1}^{n+1}\} \sim \text{Uniform}(\{v_i\}_{i=1}^{n+1}), \quad (1)$$

which implies that conditioning on the realizations of the $n + 1$ scores $\{v_i\}_{i=1}^{n+1}$, the test score $V(X_{n+1}, Y_{n+1})$ can be viewed as a draw from the uniform distribution over all the realized scores. The score function, also referred to as the *non-conformity score*, quantifies how “a typical” a

¹Note that exchangeability does not imply independence. For example, consider variables $X_i + \varepsilon$, $i \in [n]$, where $X_1, \dots, X_n$ are i.i.d. and ε is a shared additive term with $\text{Var}(\varepsilon) > 0$. Then, $X_i + \varepsilon$ ’s are exchangeable but the dependence is not zero.

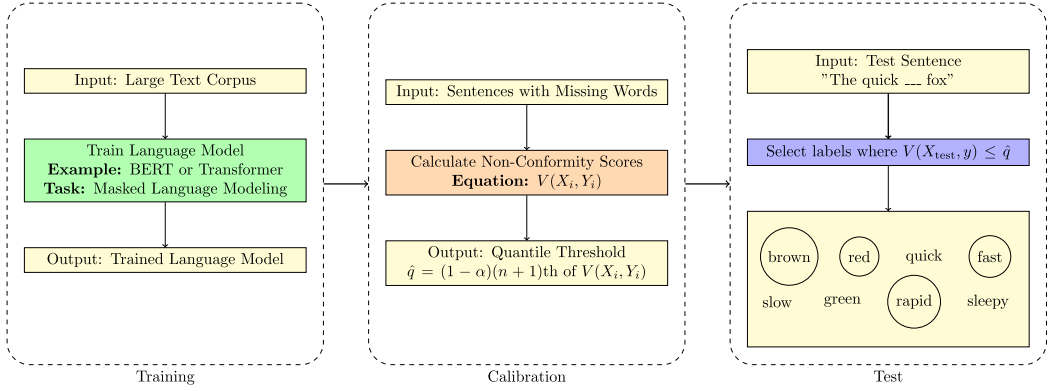


Fig. 2. Example of Split CP for Text Infilling. In this task, the model predicts missing words in a sentence, and Split CP is used to create prediction sets that quantify the uncertainty of these predictions. $V(X_i, Y_i) = 1 - P(Y_i|X_i)$, where P represents the predictive probability output by the trained base model.

prediction is relative to the true value. Then, denoting the $(1 - \alpha)$ th-quantile of the uniform distribution over $\{V(X_i, Y_i)\}_{i=1}^{n+1}$ by $Q_{1-\alpha}$, based on the property of the uniform distribution, we have

$$\mathbb{P}(V(X_{n+1}, Y_{n+1}) \leq Q_{1-\alpha}) \geq 1 - \alpha. \quad (2)$$

However, as Y_{n+1} is not observed, consider plugging in $Y_{n+1} = y$ in model fitting. Then, we denote $V^y(X, Y)$ as the score function trained with $Y_{n+1} = y$ and $Q_{1-\alpha}^y$ as the $(1 - \alpha)$ th-quantile of the uniform distribution over $\{V^y(X_{n+1}, y)\} \cup \{V^y(X_i, Y_i)\}_{i=1}^n$. Consider the following set

$$C_{1-\alpha}(X_{n+1}) = \{y \in \mathcal{Y} : V^y(X_{n+1}, y) \leq Q_{1-\alpha}^y\}, \quad (3)$$

which, according to Equation (2), satisfies that $\mathbb{P}(Y_{n+1} \in C_{1-\alpha}(X_{n+1})) = \mathbb{P}(V^{Y_{n+1}}(X_{n+1}, Y_{n+1}) \leq Q_{1-\alpha}^{Y_{n+1}}) \geq 1 - \alpha$. This indicates that with a probability no less than $1 - \alpha$, the response Y_{n+1} is included in the constructed set. A common choice of the score function is $V^y(X, Y) = |Y - f^y(X)|$, where $f^y(X)$ is some fitted estimate of $\mathbb{E}[Y | X]$ trained with $Y_{n+1} = y$. In Figure 2, this process is demonstrated for a text infilling task without calibration. Here, the score function measures the model's uncertainty in predicting missing words on training data itself, and the resulting prediction set captures multiple plausible completions while maintaining the required coverage guarantee.

2.2 Split CP and Variants

However, as we may have seen from above, full CP generally requires model re-fitting for all the values $y \in \mathcal{Y}$, which is infeasible in general scenarios with $|\mathcal{Y}| = \infty$. For computational efficiency, various variants of full CP are proposed, as detailed below.

Split CP One of the most common approaches in CP is *split CP* [105, 139], which begins by pre-fitting a model f and subsequently assessing its efficacy using a separate calibration set of labeled examples $\{(X_i, Y_i)\}_{i=1}^n$, with which we obtain non-conformity scores $\{V_i = V(X_i, Y_i)\}_{i=1}^n$.

Central to this method is the foundational assumption of exchangeability and the independence between the training and calibration sets. Based on the scores, we can construct the prediction set

$$C(X_{n+1}) = \{y \in \mathcal{Y} : V(X_{n+1}, y) \leq \widehat{Q}_{1-\alpha}\}, \quad (4)$$

where the quantile $\widehat{Q}_{1-\alpha}$ can be directly obtained from the non-conformity scores:

$$\widehat{Q}_{1-\alpha} = \text{the } \lceil (1 - \alpha)(n + 1) \rceil \text{th smallest } V_i. \quad (5)$$

The coverage guarantee of split CP ensures that the true value of a new observation falls within the predicted interval with high probability, i.e., $\mathbb{P}(Y_{n+1} \in C(X_{n+1})) \geq 1 - \alpha$. To illustrate this process, Figure 2 provides a concrete example of Split CP applied to a text infilling task. The figure highlights how the calibration process enables a trained model to compute non-conformity scores and subsequently generate prediction sets with coverage guarantees.

Split CP involves just a single model fitting step, which, while computationally efficient, can lead to lower accuracy. Due to the reduction of sample size. In contrast, full CP achieves high statistical efficiency by requiring an extensive number of model fittings [9, 105]. However, these two methods represent extremes on a spectrum of possibilities. There are intermediate techniques that balance statistical and computational efficiency in different ways.

CP with cross-validation As an extension of split CP, [184, 187] proposed cross-CP, which involves splitting the training dataset into K disjoint subsets, $S_1, \dots, S_K$, each of equal size $m = n/K$, where n is divisible by K . For each subset S_k , a regression model f_{-S_k} is trained on the complement of S_k in the dataset. These models are then used to predict the outcomes of the samples within their respective subsets, resulting in a set of cross-validation residuals, e.g.

$$V_i^{\text{CV}} = |Y_i - f_{-S_{k(i)}}(X_i)|, \quad (6)$$

where $k(i)$ identifies the subset containing instance i . Consider the text infilling task, the text corpus is split into k disjoint subsets of sentences with missing words. For each subset, a language model is trained on the other $k - 1$ subsets and then used to predict the missing words in the held-out subset. However, the finite-sample validity of the cross-conformal predictors [184, 187] at the level of $1 - \alpha$ may not hold. To address this issue, [15] proposed CV+ prediction intervals unsymmetrical around the prediction for X_{n+1} but have the finite-sample coverage at least $1 - 2\alpha$.

CP with jackknife Jackknife-based CP can be viewed as a specific example of cross-validated CP with $K = n$. Barber et al. [15] propose CP with jackknife+, which relieves the overfitting of canonical jackknife intervals and exhibits a finite-sample guarantee. Concretely, with **leave-one-out (LOO)** predictors f_{-i} , $i \in [n]$, one can calculate the residuals $V_i^{\text{LOO}} = |Y_i - f_{-i}(X_i)|$ and construct the prediction interval similar with Equation (6) by defining $k(i) = i$. In the text infilling task, Jackknife+ trains a language model by leaving out one sentence at a time, computes residuals from the model's predictions, and then uses these residuals to generate prediction sets for new inputs.

This modification is crucial in scenarios where predictions are sensitive to slight data variations, providing intervals that more accurately reflect the model's uncertainty. The Jackknife+ prediction interval satisfies the following coverage guarantee:

$$\mathbb{P}\{Y_{n+1} \in \widehat{C}_{n, 1-\alpha}^{\text{jackknife}+}(X_{n+1})\} \geq 1 - 2\alpha. \quad (7)$$

Moreover, jackknife-minimax builds on the foundation of Jackknife+ by adopting a more conservative approach, aiming to remove the factor of two from the theoretical error bound to achieve tighter coverage control as is shown in [15]. Tighter coverage guarantees around $1 - \alpha$ are also presented in [15] when algorithmic stability is assumed. Alternatives to relieve the computational costs of full CP are also explored in the literature. For example, [127] proposes approximate full CP and utilizes influence functions to construct easier-to-compute non-conformity scores.

2.3 Weighted CP

The exchangeability assumption is crucial in ensuring the finite-sample guarantee of CP. However, in many practical scenarios with population heterogeneity or distribution shift, this assumption can be violated. Tibshirani et al. [180] extended the principle of exchangeability to a weighted setting, providing a theoretical foundation for weighted CP when the data are weighted exchangeable.

More concretely, consider the setting with covariate shift:

$$(X_i, Y_i) \stackrel{\text{i.i.d.}}{\sim} P = P_X \times P_{Y|X}, \quad i \in [n], \quad (X_{n+1}, Y_{n+1}) \sim \tilde{P} = \tilde{P}_X \times P_{Y|X}. \quad (8)$$

The data are weighted exchangeable in the sense that, with the weight function $w(x) = \frac{d\tilde{P}_X}{dP_X}(x)$,

$$\left(V(X_{n+1}, Y_{n+1}) \mid \{V(X_i, Y_i)\}_{i \in [n+1]} \right) \sim \sum_{i=1}^{n+1} \frac{w(X_i)}{\sum_{j=1}^{n+1} w(X_j)} \delta_{V(X_i, Y_i)}, \quad (9)$$

where δ is the point mass function. This means that given the observed values $\{V(X_i, Y_i)\}_{i \in [n+1]}$, the new data point (X_{n+1}, Y_{n+1}) is considered to be drawn from a weighted combination of the empirical distribution of the observed data. Weighted CP has been used across various fields such as experimental design [54], survival analysis [27], and causal inference [108]. Subsequent studies have primarily focused on tackling the challenge of accurately estimating high-dimensional likelihood ratios to ensure effective coverage in complex data settings. For instance, Barber et al. [16] extend it to general non-exchangeable settings, Bhattacharyya and Barber [22] enhance coverage guarantees in scenarios where observations belong to distinct groups, and Liu et al. [116], Lu and Kalpathy-Cramer [119], Ying et al. [202] bolster the informativeness of weighted CP when facing covariate shifts across multiple data sources. To address the problem with distribution shift, there is another line of research focusing on calibrating the confidence level α to maintain validity for the worst-case distribution [1, 28, 76], which is known as distributionally robust methods.

2.4 Conformal Risk Control

Conformal risk control [10] is a more general framework compared to traditional CP. While CP focuses on constructing prediction sets with guaranteed coverage probability, conformal risk control expands this idea to allow for more flexible risk management, considering different risk functions. Given a parametrized family of prediction sets $\{C_\lambda(x) : \lambda \in \Lambda\}$, e.g., $C_\lambda(x) = \{y \in \mathcal{Y} : V(x, y) \leq \lambda\}$ with a given non-conformity score function V , the problem of constructing a valid prediction set boils down to the calibration of the parameter λ . The main idea is to choose λ such that the average miscoverage in the calibration set is controlled, i.e.,

$$\hat{\lambda} = \inf \left\{ \lambda \in \Lambda : \frac{1}{n_{\text{cal}} + 1} \sum_{i=1}^{n_{\text{cal}}} \mathbb{1}\{Y_i \notin C_\lambda(X_i)\} + \frac{1}{n_{\text{cal}} + 1} \leq \alpha \right\}. \quad (10)$$

Then, the prediction set $C_{\hat{\lambda}}(x)$ satisfies that $\mathbb{P}(Y_{n+1} \in C_{\hat{\lambda}}(X_{n+1})) \geq 1 - \alpha$. Extensions to the generic loss function $\ell(x, y)$ are also presented in [10]. Similar ideas are also explored in [141–143], in which PAC-type control of coverage is established.

2.5 Evaluation Metrics

In this section, we present a summary of common validity and accuracy guarantees for CP under a supervised setting. We denote $\hat{C}_{n, 1-\alpha}$ as the prediction set based on the previous n samples $\{(X_i, Y_i)\}_{i \leq n}$ at the level of $1 - \alpha$ and consider the following metrics.

2.5.1 Marginal Coverage. The most basic guarantee for a prediction interval is the finite-sample marginal coverage such that for the test sample X_{n+1} , one has

$$\mathbb{P}(Y_{n+1} \in \hat{C}_{n, 1-\alpha}) \geq 1 - \alpha, \quad (11)$$

where the probability is with respect to the randomness of $\{(X_i, Y_i)\}_{i \leq n+1}$. Furthermore, to obtain tight coverage, it usually requires that $|\mathbb{P}(Y_{n+1} \in \hat{C}_{n, 1-\alpha}) - (1 - \alpha)| = O(n^{-1})$. CP is shown to meet both requirements with exchangeable data [3, 106, 162, 186].

2.5.2 Adversarial and Instantaneous Coverage in Online Setting. In the online setting with spatio-temporal data, where data arrive sequentially and are not guaranteed to have the same distribution (see more details in Section 5), the exchangeability assumption is violated, thus the worst-case or adversarial coverage is considered in the literature.

Adversarial marginal coverage. The worst-case coverage is the *mean prediction interval coverage* averaged over all time points, which can be written as $T^{-1} \sum_{t=1}^T \mathbb{1}\{Y_t \in \widehat{C}_t(X_t)\}$, where T is the total number of time points and $\widehat{C}_t(\cdot)$ is the prediction interval constructed at time t , and 0 otherwise [16, 21, 68, 88, 112, 166, 169, 174, 192, 194, 200, 204].

Instantaneous coverage. The worst-case marginal validity holds without distributional assumption for **Adaptive conformal Inference (ACI)**-type approaches, or alternatively, can be viewed as the property of a fixed sequence. To show the adaptivity of online CP approaches, [7] also validates that the ACI-type has instantaneous validity under the i.i.d. assumption with sufficiently large t . With a fitted non-conformity score s_t , [7] consider the asymptotic control of the coverage rate at time t : $\text{Coverage}_t(q) = \mathbb{P}(V_t(X, Y) \leq q \mid V_t)$.

2.5.3 Conditional Coverage. Conditional coverage is a stronger validity criterion. For example, if we are interested in a subpopulation determined by a function $g(x, y)$, it leads us to prediction intervals satisfying the conditional coverage guarantee such that for all $\widetilde{g}$,

$$\mathbb{P}\left(Y_{n+1} \in \widehat{C}_{n,1-\alpha}(X_{n+1}) \mid g(X_{n+1}, Y_{n+1}) = \widetilde{g}\right) \geq 1 - \alpha. \quad (12)$$

In the following subsections, we consider different choices of the function g in practice.

Covariate-conditional coverage. A natural partition of the entire space is according to the realization of the test feature, i.e., $g(x, y) = x$, in which case we seek the covariate-conditional (or local) coverage. Unfortunately, when the marginal distribution of X has no point mass, this guarantee is shown to be impossible in [107, 186]. Various kinds of relaxation of the exact conditional coverage are explored in the literature (see [62, 73, 84, 105, 152]).

Class-conditional coverage. In the task of classification, it is of interest to provide valid prediction interval conditioning on each class, i.e., $g(x, y) = y$, in which we consider the label conditional guarantee. In addition to the Mondrian CP in [45, 186] archives class-conditional validity in the setting where the number of classes is large such that effective samples in each class may not be sufficient. Empirical studies are conducted in [74, 82, 117, 156, 164, 172].

Training-conditional coverage. Since the coverage probably has high variability in the training data, it drives us to consider the coverage guarantee conditional on the training set $\mathcal{D}_n$:

$$\mathbb{P}(\alpha(\mathcal{D}_n) \geq \alpha + \varepsilon_n) \leq \delta_n, \text{ where } \alpha(\mathcal{D}_n) = \mathbb{P}\left(Y_{n+1} \notin \widehat{C}_{n,1-\alpha}(X_{n+1}) \mid \mathcal{D}_n\right), \quad (13)$$

and $\varepsilon_n, \delta_n \rightarrow 0$ as $n \rightarrow +\infty$. Specifically, the marginal coverage can be written as $\mathbb{E}[\alpha(\mathcal{D}_n)] \leq \alpha$.

2.5.4 Prediction Efficiency. The efficiency of the prediction set/interval measures its size or length $\|\widehat{C}_{n,1-\alpha}\|$. It is also common to compare with an oracle prediction set $C_{1-\alpha}^*$ under some specified models, see e.g., [105]. More criteria to measure prediction efficiency are discussed in [185]. For example, one can evaluate the second-highest p-values or the sum of all p-values excluding the largest one, as these metrics reflect the distribution of confidence across potential labels. In the online setting, the mean prediction interval width, rolling width, and average median length of the prediction intervals are considered. To address the issue of infinite-length intervals, it is suggested to measure the *fraction of infinite-length prediction intervals* [200] and analyze the 75%, 90%, and 95% quantiles of the set sizes [6] to understand the distribution of interval lengths better.

2.5.5 Hybrid Metric. There are also several metrics that consider validity and efficiency together. [113] introduces the *inverse coverage efficiency*, which is the average prediction interval length divided by the marginal coverage rate (the smaller, the better). [88] considers the **coverage width-based criterion (CWC)** that penalizes under- and overcoverage in a similar way. Additional metrics that aim at balancing validity and accuracy are explored in [14, 166].

3 Static Data-Structured Data

3.1 Flat Data

We start with the scenario where data are numeric and assumed to be independently and identically distributed (i.i.d or *flat*), which is common in regression/classification problems.

3.1.1 Regression. Consider observed covariates $\{X_i\}_{i=1}^n \subseteq \mathbb{R}^d$ and response $\{Y_i\}_{i=1}^n \subseteq \mathbb{R}$, where we assume $\{(X_i, Y_i)\}_{i=1}^n$ are i.i.d. Then, according to the previous section, for any non-conformity score $V(x, y)$, the prediction set defined in Equation (4) is valid. Particularly, for the regression problem, the score $V(x, y) = |y - \hat{f}(x)|$ is widely adopted [105, 162, 186] with a fitted function f , with which the prediction interval takes the form $C_{1-\alpha}(X_{n+1}) = f(X_{n+1}) \pm \hat{Q}_{1-\alpha}$. Note that the prediction interval has an equalized width for different X_{n+1} . To incorporate the local uncertainty of covariates when the noise distributions are heteroscedastic, a locally weighted score is proposed [105, 139]:

$$V(x, y) = \frac{|y - \hat{f}(x)|}{\hat{\sigma}(x)}, \quad (14)$$

where the local uncertainty score $\hat{\sigma}(x) > 0$ is an estimated standard error of the residual. Variants of the local uncertainty score are proposed and applied to various scenarios [39, 83, 138].

However, as the distribution of $(Y | X)$ may not lie in the location-scale family such that $Y = f(X) + \sigma(X) \cdot e$, the prediction interval based on the locally weighted score can lose efficiency when dealing with more complicated distributions. Another notable line of research focuses on non-conformity scores based on more general distributional quantities such as estimated quantiles and **cumulative density functions (CDF)**, which are introduced as follows.

Conformalized quantile regression (CQR). Even when the distribution does not belong to the location-scale family, the quantile function can capture the intrinsic characteristics (e.g., the shape) of the distribution. [152] proposes CQR where the prediction interval is defined as

$$\left[\hat{Q}_{\alpha/2}(X_{n+1}) - \eta_{1-\alpha}, \hat{Q}_{1-\alpha/2}(X_{n+1}) + \eta_{1-\alpha} \right], \quad (15)$$

where $\hat{Q}$ is the estimated quantile based on data splitting and on the calibration set with sample size n_{cal} , $\eta_{1-\alpha}$ is the $(1 - \alpha)(1 + 1/n_{\text{cal}})$ -th-quantile of

$$V_i = \max \left\{ \hat{Q}_{\alpha/2}(X_i) - Y_i, Y_i - \hat{Q}_{1-\alpha/2}(X_i) \right\}, \quad 1 \leq i \leq n_{\text{cal}}. \quad (16)$$

When the estimated quantile is accurate, CQR also exhibits good local performance and to improve local coverage, further extensions are proposed in [4, 154, 160, 167].

Distributional CP. In comparison, with the same motivation to capture the characteristics of the distribution beyond the location-scale family and achieve better conditional performance, distributional CP [35] considers the scores based on an estimated CDF. Consider the data splitting case where $\hat{F}(x, y)$ is an estimated CDF using a hold-out set. [35] defines the scores as

$$V_i = \psi \left(\hat{F}(X_i, Y_i) \right), \quad 1 \leq i \leq n_{\text{cal}}, \quad (17)$$

where ψ is a user-specified scalar function to enhance accuracy and the default choice is $\psi(t) = |t - 1/2|$. When the CDF is estimated accurately, distributional CP is also shown to have an asymptotic conditional coverage guarantee.

Discussion. CP with flat data in the regression setting mainly focuses on designing non-conformity scores to achieve better local coverage. Further discussion is presented in Section 2.5.3.

3.1.2 Classification. To apply CP to classification problems with K classes such that $Y \in [K]$, a widely adopted non-conformity score is the estimated likelihood [156, 162]. Taking split CP as an example where we have $\widehat{p}(x; k)$ as an estimate for $\mathbb{P}(Y = k \mid X = x)$, then with the non-conformity score $V_i = \widehat{p}(X_i; Y_i)$, the prediction set takes the form

$$C_{1-\alpha}(X_{n+1}) = \left\{ k : \widehat{p}(X_{n+1}; k) \geq \widehat{Q}_{1-\alpha} \right\}, \quad (18)$$

where $\widehat{Q}_{1-\alpha}$ is the $\lfloor \alpha \cdot (1 + n_{\text{cal}}) \rfloor$ -th-smallest of $\{V_i\}_{i=1}^{n_{\text{cal}}}$.

A modified approach is proposed in [153] where the non-conformity score takes the form

$$\widetilde{V}_i = \sum_{k \in S_i} \widehat{p}(X_i; k), \quad S_i = \{k : \widehat{p}(X_i; k) \geq \widehat{p}(X_i; Y_i)\}, \quad (19)$$

which is a score that tends to be small for the more “likely” labels for a certain x . Let $\widehat{Q}_{1-\alpha}$ be the $\lfloor (1 - \alpha)(1 + n_{\text{cal}}) \rfloor$ -th-smallest of $\{\widetilde{V}_i\}_{i=1}^{n_{\text{cal}}}$ and $\{\pi(j; x)\}_{j=1}^K$ be permutation of $[K]$ such that

$$\widehat{p}(x; \pi(1; x)) \geq \widehat{p}(x; \pi(2; x)) \geq \cdots \geq \widehat{p}(x; \pi(K; x)). \quad (20)$$

Then, the prediction set takes the form

$$C_{1-\alpha}(X_{n+1}) = \{\pi(k; X_{n+1}) : 1 \leq k \leq k(X_{n+1})\}, \quad (21)$$

where $k(x) = \min\{k : \sum_{j=1}^k \widehat{p}(x; \pi(j; x)) \leq \widehat{Q}_{1-\alpha}\}$ and $\widehat{Q}_{1-\alpha}$ is the $\lfloor (1 - \alpha)(1 + n_{\text{cal}}) \rfloor$ -th-smallest of $\{\widetilde{V}_i\}_{i=1}^{n_{\text{cal}}}$. In [29], a non-conformity score based on estimated quantiles, which is similar to CQR, is adopted for better local coverage performance. More recent work [85] further improves the proposed non-conformity score via label ranking. [52] calibrates the output probability of deep neural networks for more informative prediction sets.

Ordinal classification. When the labels are ordinal, the prediction set is required to be contiguous. However, the prediction sets constructed above for general classification problems do not have this property unless the estimated conditional distribution $\widehat{P}_{Y|X}$ is unimodal. Motivated by this property, [43] proposed a procedure that constructs unimodal classifiers. In addition, [30] adopts conformal p-values to determine the lower and upper bounds for the contiguous prediction set for the ordinal responses, and [198] applies the conformal risk control framework to this setting. Conformalized ordinal classification has applications in disease severity rating [30, 43, 118], age recognition [43, 198], and historical image dating [43], and so on.

Multi-label classification. Maxime Cauchois et al. [29] apply CP on multi-class and multi-label image classification. One key challenge here is that typical CP methods, which give marginal validity (coverage) guarantees—provide uneven coverage, in that they address easy examples at the expense of essentially ignoring difficult examples. They build tree-structured classifiers that efficiently account for interactions between labels. To asymptotically get conditional coverage for both multi-class and multi-label prediction, they adopt the idea from quantile regression [152].

Discussion. Conditional coverage is also of concern in classification problems, which motivates a series of works [29, 153] in constructing scores similar to conditional quantiles. In addition, UQ for ordinal classification with additional rank constraints on the labels, which lies between regression and classification with unordered labels, is still an active field. Selective classification

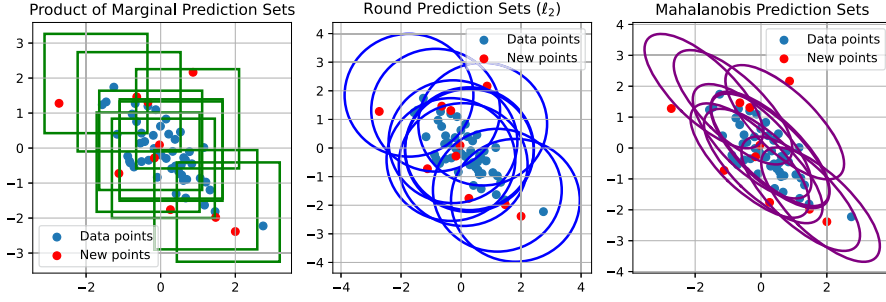


Fig. 3. Illustration of **conformal prediction (CP)** sets with bivariate response.

is also of interest in ML, where more cautious prediction sets with the ability to abstain are constructed [82]

3.1.3 Multivariate Response. As an extension to CP with univariate response Y , the setting with a multivariate response is also of interest in the scenario where $Y \in \mathbb{R}^k$ is a k -dimensional series for each data point. Although the validity of CP still holds with properly defined non-conformity scores, the specification of the scores is a crucial question to determine the shape of the prediction set for better accuracy. A naive approach is to construct the CP prediction interval $\widehat{C}_j(X_{n+1})$ at the level of $1 - \alpha_j$ for $Y_{n+1,j}$ for each $j \in [k]$ as is depicted in Figure 3 (left panel). Then, by the union bound, one has $\mathbb{P}\{Y_{n+1} \in \widehat{C}_1(X_{n+1}) \times \cdots \times \widehat{C}_k(X_{n+1})\} \geq 1 - \sum_{j=1}^k \alpha_j$. Specifically, when coordinates in Y are independent, we can further have $\mathbb{P}\{Y_{n+1} \in \widehat{C}_1(X_{n+1}) \times \cdots \times \widehat{C}_k(X_{n+1})\} \geq \prod_{j=1}^k (1 - \alpha_j)$, which has much tighter coverage. To achieve a middle-ground between the independent and adversarially dependent settings, [129] and [174] further consider the copula transformation to achieve independent coordinates and apply an inverse map to obtain the predict set for the multivariate response. In comparison, [95] extends the score $|Y - \widehat{f}(X)|$ to $\|Y - \widehat{f}(X)\|$ with different choices of the norm $\|\cdot\|$ in $\mathbb{R}^k$ and define the prediction set by $\{y \in \mathbb{R}^k : \|y - \widehat{f}(X_{n+1})\| \leq \widehat{q}_{1-\alpha}\}$ with calibrated quantile $\widehat{q}_{1-\alpha}$ for the scores. See Figure 3 (middle and right panels) for prediction sets with two different choices of norms: $\|Y - \widehat{f}(X)\|_2$ and $\|Y - \widehat{f}(X)\|_{\Omega} = (Y - \widehat{f}(X))^T \Omega^{-1} (Y - \widehat{f}(X))$, where Ω is an estimate of the covariance matrix of residual vectors. Other extensions to the setting with multivariate responses are discussed in [44, 47, 56, 192].

3.1.4 Functional Data. CP with functional data can be viewed as a further extension to the previous section with multivariate responses since the response here is a function in an infinite-dimensional space. Consider the setting with random functions $X_1(\cdot), \dots, X_n(\cdot) \in \Omega$ with domain $\mathcal{T}$. The goal is to find a prediction set $C_{1-\alpha} \subseteq \Omega$ such that $\mathbb{P}(X_{n+1} \in C_{1-\alpha}) \geq 1 - \alpha$. For brevity, we consider the split CP. To solve this problem, a naive trial is to find a predefined function $g : \Omega \rightarrow \mathbb{R}$, e.g., $g(X) = -\sup_{t \in \mathcal{T}} |X(t)|$, and then define $C_{1-\alpha} = \{x \in \Omega : g(x) \geq \lambda\}$. Here λ is $(\lceil \alpha(n_{\text{cal}} + 1) \rceil - 1)$ -th-smallest among $\{g(X_i)\}_{i=1}^{n_{\text{cal}}}$. However, this criterion is too ambitious and may lose efficiency. In [106], a modified criterion is considered: $\mathbb{P}(\Pi(X_{n+1}) \in C_{1-\alpha}) \geq 1 - \alpha$, where Π maps functions in Ω to a finite dimensional space $\Omega_p \subset \Omega$. More concretely, [106] considers the prediction bands $B(t) = [l(t), u(t)]$, $t \in \mathcal{T}$, such that $\mathbb{P}(l(t) \leq X_{n+1}(t) \leq u(t), \forall t \in \mathcal{T}) \geq 1 - \alpha$. The proposed approach projects functions to vectors $\{\xi_i\}_{i \leq n} \subseteq \mathbb{R}^p$ in the Euclidean space via a basis $\{\phi_j\}_{j=1}^p$. Then, with non-conformity scores $V(\xi_i)$, $i \in [n_{\text{cal}}]$ and

$$\mathcal{E}_n = \{\zeta \in \mathbb{R}^p : V(\zeta) \geq V_{(\lceil \alpha(n_{\text{cal}}+1) \rceil - 1)}\}, \quad (22)$$

where $V_{(k)}$ is the k th order statistic of $\{V(\xi_i)\}_{i=1}^{n_{\text{cal}}}$, one can construct the prediction band by $B(t) = \{\sum_{j=1}^p \zeta_j \phi_j : \zeta \in \mathcal{E}_n\}$. In practice, the non-conformity score $V(\cdot)$ is chosen to be the Gaussian mixture density estimator with K components. [46] constructs prediction bands for X_{n+1} without projection by considering the non-conformity score

$$V(X_i) = \sup_{t \in \mathcal{T}} \left| \frac{X_i(t) - \widehat{G}(t)}{\widehat{s}(t)} \right|, \quad i = 1, \dots, n_{\text{cal}}, \quad (23)$$

where $\widehat{G}(t)$ is the mean of $X_i(t)$ in the hold-out training set and $\widehat{s}(t)$ is designed to penalize extreme values in [46]. This procedure is further extended to the multivariate functional setting [47].

3.1.5 Data with Noise, Missingness, or Censoring. The ideal exchangeability assumption can be violated in many scenarios, among which data corruption is a common setting in statistics. In this section, we summarize related works on CP with flat data but either the covariates or the response is not directly observed, which can be missing or perturbed.

Covariate noise. The simplest setting with data corruption is the case with noisy covariates for the test data, which may happen when measurement errors are taken into account [38] or perturbations are applied to enhance privacy [51]. Consider the case where we observe $\{(X_i, Y_i)\}_{i=1}^n$ and a new sample with noisy covariates $\tilde{X}_{n+1}$. Gendler et al. [65] adopt randomized smoothing to construct a non-conformity score $\tilde{V}(x, y)$ which satisfies that

$$V(\tilde{X}_{n+1}, y) \leq V(X_{n+1}, y) + M_\delta, \quad \forall y \in \mathcal{Y}, \quad \text{almost surely}, \quad (24)$$

where M_δ is a constant. Concretely, given any $V(x, y)$, the score $\tilde{V}(x, y)$ can be constructed by

$$\tilde{V}(x, y) = \Phi^{-1} \left(\mathbb{E}_{v \sim \mathcal{N}(0_d, I_d)} [V(x + v, y)] \right), \quad (25)$$

where Φ is the CDF of $\mathcal{N}(0_d, I_d)$. A more recent work [67] only requires $V(\tilde{X}_{n+1}, y) \leq V(X_{n+1}, y) + M_\delta$ to hold with a high probability and proposes an adaptive approach to achieve improved trade-offs between performance on clean data and robustness to adversarial examples based on the principle of quantile-of-quantile design. [205] further considers the performance of CP under poisoning/evasion attack.

Label noise. [57] focuses on the setting where we observe $\{(X_i, g(Y_i))\}_{i=1}^n$ with a corruption function $g(\cdot)$ and the goal is to construct a valid prediction interval for Y_{n+1} . For any $u \in (0, \alpha - 1/(n+1))$, if the test score $V_{n+1} = V(X_{n+1}, Y_{n+1})$ and the corrupted score $\tilde{V}_{n+1} = V(X_{n+1}, \tilde{Y}_{n+1})$ satisfy

$$\mathbb{P}(\tilde{V}_{n+1} \leq t) + u \leq \mathbb{P}(V_{n+1} \leq t), \quad (26)$$

then with the protection interval $\tilde{C}_{1-\alpha}(\cdot)$ based on $(X_i, g(Y_i))$'s, it holds that

$$\mathbb{P}(Y_{n+1} \in \tilde{C}_{1-\alpha}(X_{n+1})) \geq 1 - \alpha + \frac{1}{n+1} + u. \quad (27)$$

Moreover, if the noisy distribution of $\tilde{Y} = g(Y)$ satisfies that $d_{\text{TV}}(Y, \tilde{Y}) \leq \varepsilon$, one has

$$\mathbb{P}(Y_{n+1} \in \tilde{C}_{1-\alpha}(X_{n+1})) \geq 1 - \alpha + \frac{1}{n+1} + \frac{n}{n+1} \varepsilon, \quad (28)$$

which shows CP sets are robust when the strength of label noise is mild. In the study [161], they propose a novel CP method that adjusts for label contamination through a precise characterization of the effective coverage inflation or deflation caused by label contamination. Recently, a non-conformity score that interpolates noisy and clean data to achieve robustness is proposed [144].

Covariate missingness. In [203], the setting with missing coordinates in X_i 's is considered, which can be viewed as a specific case of noisy covariates. For example, the test data can take the form $X_{n+1} = (X_{n+1,1}, \cdot, \cdot, X_{n+1,4})$, where the second and third entries are unobserved. Solutions

are proposed in [203] and the main idea is to artificially create missingness in the calibration dataset such that the second and third entries in calibration data are also missing. In the first approach, calibration data with additional missingness (e.g., $X_j = (X_{j,1}, \cdot, \cdot, \cdot)$) will be removed, in which case the exchangeability is recovered. To relieve the issue of sample size insufficiency, [203] also suggests that we could impute X_j to obtain $X_j = (X_{j,1}, \cdot, \cdot, \hat{X}_{j,4})$ for CP. Another line of work focusing on data (e.g., matrix or tensor) completion will be introduced in Section 3.2.

Causal inference. Causal inference under the potential outcome model [155] can be viewed as an example of the missing response. Consider covariates X_i 's and potential outcomes $(Y_i(0), Y_i(1))$ for each unit. A treatment $T_i \in \{0, 1\}$ is assigned to unit i and the strong ignorability assumption is commonly considered: $(Y_i(0), Y_i(1)) \perp\!\!\!\perp T_i \mid X_i$, i.e., no unobserved hidden confounders. The two major challenges of applying CP in causal inference are the covariate shift and inductive bias due to counterfactuals. In [108], under the strong ignorability assumption, prediction intervals are constructed for random variables, such as $Y_{n+1}(1)$ and $Y_{n+1}(1) - Y_{n+1}(0)$, via weighted CP. To see this, for $Y_{n+1}(1)$, the density ratio between the population of interest $P_{Y(1), X|T=1}$ and $P_{Y(1), X}$ only depends on X , which lies in the covariate shift setting. Different choices of weight functions are presented in [108] for various inferential targets. In comparison, [98] adopts CP without reweighting to obtain prediction intervals that are valid conditioning on the treatment T . [33, 89] propose robust CP that maintains valid coverage when the assumption is slightly violated. More concretely, for example, with the bounds constraint of the density ratios, the weight function can be replaced by a lower or an upper bound function in calculating the weighted quantile for weighted CP. In addition, [32, 146, 199] further propose doubly robust results by incorporating the missingness/covariate shift in the framework of non-parametric/semi-parametric statistics and Alaa et al. [3] propose conformal meta learner which combines CP with meta-learner [100] to estimate causal effects.

Censored data. The setting with the censored response is commonly known as survival analysis where X consists of covariates, T is the survival time of interest and C is the censoring time. In practice, only the censored time $\tilde{T} = \min\{C, T\}$ can be observed. With the goal of constructing a **lower prediction bound (LPB)** for T such that $\mathbb{P}(L_{1-\alpha}(X) \leq T) \geq 1 - \alpha$, if we directly treat $\tilde{T}$ as the response, the LPB for $\tilde{T}$ is also valid for T . However, when $C < T$ with a high probability, the LPB for $\tilde{T}$ will be too conservative for T . The main idea in [27, 77] is to find a cutoff $q(X)$ for C and focus on a subpopulation with $C \geq q(X)$, in which the censoring rate is lower. The difference lies in that [27] adopts a fixed cutoff c_0 while [77] considers an adaptive cutoff as a function of X . We note that, under the conditionally independent censoring assumption that $T \perp\!\!\!\perp C \mid X$, the distribution shift from (T, C, X) to $(T, C, X \mid C \geq q(X))$ is covariate shift. Thus, by weighted CP, the LPB for $\tilde{T}$ constrained on the subpopulation will gain more accuracy.

Discussion. UQ for a wide range of statistical problems with missing/noisy observations can be addressed via (weighted) CP. However, its validity usually depends on the covariate shift condition, which has different variants in specific problems (e.g., the strong ignorability assumption in causal inference and the conditionally independent censoring assumption in survival analysis). Motivated by [16, 89], there is ongoing interest in robustness evaluation or sensitive analysis of (weighted) CP when the covariate shift assumption is violated.

3.2 Other Structured Data

The validity of CP holds with flat data. However, in practice, the i.i.d. assumption is too strong and does not hold in general. This section provides an overview of applications of CP with hierarchical, matrix/tensor, and graph data and additional assumptions required for the exchangeability to hold.

3.2.1 Data with Hierarchical Structure. When data are generated via a hierarchical sampling procedure, it creates a simple form of heterogeneity within data. Consider the setting with K many

groups of observations and $\Pi_k \sim P_\Pi$, $N_k \mid \Pi_k \sim P_{N|\Pi}$, independently for $k \in [K]$, where N_k is the observed sample size for the k th group. For each group k , we have observations $\{(X_{k,i}, Y_{k,i})\}_{i=1}^{N_k}$. As mentioned in [49], there are two tasks of interest: (1) constructing a prediction set for the new sample $(X_{K+1,1}, Y_{K+1,1})$ drawn from a new group $\Pi_{K+1} \sim P_\Pi$; (2) constructing a prediction set for the new sample $(X_{\text{test}}, Y_{\text{test}})$ from one of existing K groups. Let $\tilde{Z}_k = (Z_{k,i})_{i=1}^{N_k} = ((X_{k,i}, Y_{k,i}))_{i=1}^{N_k}$ for $k \in [K+1]$. As shown in [49], in this setting, the hierarchical exchangeability condition is assumed as an extension of the generic exchangeability assumption, where not only are observations within each group exchangeable, K groups are also assumed to be exchangeable.

For task (1), four methods are proposed in [49]. The first approach, *pooled CDF*, utilizes the estimated CDF from each group to construct asymptotically valid prediction sets. The second approach, *double conformal*, is similar to the quantile-of-quantile approach and achieves finite-sample validity under the hierarchical exchangeability assumption but tends to be conservative in practice. The third approach, *subsampling once*, samples one sample from each group and constructs the prediction set following the generic CP procedure without the requirement of hierarchical exchangeability. The shortcoming is that this approach does not utilize all the available data and may exhibit high variance. In light of this, *repeated subsampling* repeats the procedure B times to derandomize the subsampling procedure via p-values aggregation.

Another work [22] considers the setting with training and test data drawn from different multinomial distributions, and to account for the heterogeneity in the sampling probabilities, weighted CP is adopted to adapt the test sample to “similar” observations.

3.2.2 Matrix/tensor Data. An important line of research in unsupervised learning focuses on matrix estimation/prediction, e.g., forecasting in panel data with spatio-temporal correlations. In this setting, we have an underlying matrix $M \in \mathbb{R}^{d_1 \times d_2}$ of interest, and each entry (i, j) is missing following a certain pattern. With the random set of observed items $S \subseteq [d_1] \times [d_2]$, the task is to construct intervals $\{C(i^*, j^*) : (i^*, j^*) \in S^c\}$ such that $M_{i^*j^*} \in C(i^*, j^*)$ with a high probability.

In [75], it is assumed that the matrix M is deterministic and each entry (i, j) is observed with probability $p_{ij} \in (0, 1)$ independently and weighted exchangeability of indices in $S_{\text{cal}} \cup \{(i^*, j^*)\}$ conditioning on $S \setminus S_{\text{cal}}$ is established for (i^*, j^*) uniformly drawn from S^c , which enables the splitting approach that uses $S \setminus S_{\text{cal}}$ for estimating quantities of interest. More concretely, with $\hat{M}$, $\hat{p}$, and the local uncertainty score $\hat{s}$ estimated with $M_{S \setminus S_{\text{cal}}}$, we have $V_{ij} = (|M_{ij} - \hat{M}_{ij}|) / \hat{s}_{ij}$ for all $(i, j) \in S_{\text{cal}}$. Then, by weighted CP with $\hat{w}_{ij}$ as the normalized odds ratio, the prediction interval for each (i^*, j^*) can be constructed by $C(i^*, j^*) = \hat{M}_{i^*j^*} \pm \hat{s}_{i^*j^*} \hat{Q}$ where

$$\hat{Q} = \text{Quantile}_{1-\alpha} \left(\sum_{(i,j) \in S_{\text{cal}}} \hat{w}_{ij} \delta_{V_{ij}} + \hat{w}_{i^*j^*} \delta_{+\infty} \right). \quad (29)$$

This approach archives the validity in S^c on average such that

$$\frac{1}{|S^c|} \sum_{(i^*, j^*) \in S^c} \mathbb{1} \{M_{i^*j^*} \in C(i^*, j^*)\} \geq 1 - \alpha - \Delta, \quad (30)$$

where Δ measures the estimation accuracy of $\hat{p}$.

In comparison, [163] considers the setting with a random matrix and fixed missing patterns. Valid intervals are constructed under the row/column exchangeability assumption on the random matrix. Since the coverage guarantee in [75] is on average for S^c , a recent work [110] considers a different criterion of coverage, *simutaneous coverage*, which constructs a joint confidence region for M . [171] further extends the problem to the scenarios with a noisy and partially observed tensor.

3.2.3 Data with Graph (Tree) Structure. Graph data is widely used across various fields, such as recommendation systems [81] and drug discovery [63]. CP for graph data presents both unique challenges and opportunities. One key challenge arises from the reliance of standard CP on the exchangeability assumption, which may not hold for graph data due to its inherent dependencies. Several studies [36, 86, 207] have explored the conditions under which the exchangeability assumption applies to graph data. A common consensus is that in inductive settings, where test data nodes/links are not present in the calibration data, the exchangeability assumption typically does not hold. However, in transductive settings, where all nodes from both the training and testing datasets are included in the graph, the assumption is more likely to be valid. Despite this challenge, the unique structure of graph data also presents opportunities to enhance the performance of CP methods, offering untapped potential for improvement.

Inductive setting In the inductive setting, models are trained on one graph (or a subset of nodes and edges) and then applied to unseen data, such as new nodes or entirely new graphs, during inference. As a result, the exchangeability assumption does not hold, causing CP to fail to provide coverage guarantees. To address this challenge, Clarkson [36] propose the Neighbourhood Adaptive Prediction Set. By leveraging the adjacency matrix, they generate prediction sets for test data, using the neighborhoods of test nodes as calibration data and introduce a weighting scheme based on the path length between nodes to adjust the importance of nodes in calibration, inspired by the homophily principle. In a different approach, Zargarbashi and Bojchevski [207] note that the previous method may not work well with realistic sparse datasets, such as social network interactions. To tackle this issue, they suggest recalculating the non-conformity score and quantile each time new nodes are added to the graph. For edge classification, the same strategy is employed, but with a weighted non-conformity score defined as $1/\text{degree}(v)$, where v represents a node, to account for covariate shift. In Calissano et al. [25], they proposed a CP method applicable to both labeled and unlabeled graphs. For unlabeled graphs, they embed the prediction set into a quotient space, where isomorphic graphs are grouped into equivalence classes. As a result, this approach effectively handles graph structures without reliance on specific node labels.

Transductive setting In this setting, research has shown that exchangeability holds for both node and link prediction tasks [86, 211]. As a result, most works focus on improving efficiency. In [206], the authors update the non-conformity score by incorporating the non-conformity scores of neighboring nodes, effectively leveraging the graph structure to improve prediction efficiency. Specifically, they propose a diffused score as the non-conformity score:

$$\widehat{V}(v, y) = (1 - \lambda)V(x, y) + \frac{\lambda}{|\mathcal{N}_x|} \sum_{u \in \mathcal{N}_x} V(u, y), \quad (31)$$

where $\mathcal{N}_x$ is the set of neighboring nodes of node v , and λ is a hyperparameter. Huang et al. [86] further propose a more general method, highlighting that current approaches mainly focus on designing non-conformity scores while overlooking the direct training for efficiency. Based on the empirical finding that nodes linked together often produce similar sizes of prediction sets, they employ a second GNN model to help aggregate neighborhood information and introduce an inefficiency loss, defined as the size of the prediction set.

In addition to the aforementioned node classification tasks, there are some works that focus on applications of CP on link prediction. For example, extending previous exchangeability results in [86], Zhao et al. [211] prove that the assumption also holds for link prediction tasks in GNNs, and introduce conformalized link prediction. The key finding is that graphs with node degrees following the power law distribution more closely tend to display smaller average prediction intervals, indicating higher efficiency. They then propose a sampling-based CP method to align the node degree distribution of the graph with a power-law distribution to enhance the efficiency of

the CP process. Another study [126] develops a conformal link prediction method that provides a set of true edges while controlling the **False Discovery Rate (FDR)**. The motivation is that most methods consider link prediction as a ranking problem, which is satisfying when the application constrains the number of pairs of nodes to be declared as true edges to be fixed. Alternatively, other practices may require a classification of missing links into true and false edges with a control of false positives, e.g., reconstruction of biological networks. They collect a set of observed false links (those confirmed to not exist) and calculate p-values on these. Then, they apply the Benjamini-Hochberg procedure to the conformal p-values to establish a threshold for determining whether a link is a true connection. In a follow-up study, Blanchard et al. [23] validate the effectiveness of this approach and extend it to include False Discovery Proportion bounds.

Another area of link prediction is recommendation systems, which aim to predict interactions between users and items. For example, Angelopoulos et al. [12] focus on controlling FDR for evaluating variable-sized sets in drug recommendation. To achieve this, they propose a method that calibrates **learning-to-rank (L2R)** models to maintain FDR control. Additionally, in [188], the authors introduce two novel CP-based losses: **CP Set Size (CPS)** and **CP Set Distance (CPD)**. CPS is designed to improve the efficiency of CP, while CPD minimizes the cosine similarity distance between the top-K closest items in the CP set and the ground truth. This enhances recommendation precision by ensuring that items are not only selected with confidence (as with CPS) but also closely match user preferences. They train the model using joint learning to achieve these goals.

Discussion CP for graphs is a relatively underexplored area, and several challenges remain. First, the graph structure itself can be more thoroughly leveraged. Current research primarily focuses on neighborhood information, but other relationships—such as nodes with similar degrees or shared topological properties—could also be incorporated into CP. Second, there is limited research on applying CP to weighted graphs, such as transportation networks, or other graph variants, which presents a significant gap in the literature. Additionally, optimizing the efficiency of CP in settings beyond the current focus remains an open area for exploration.

4 Static Data–Unstructured Data

4.1 Text Data

The rapid expansion of **large language models (LLMs)** and NLP applications has highlighted the critical need for UQ to mitigate risks such as hallucinations and to improve decision-making reliability in high-stakes scenarios. Text and LLMs possess unique characteristics that

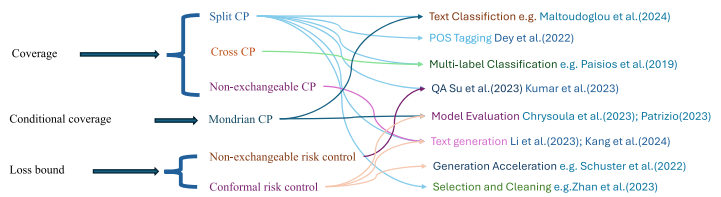


Fig. 4. A taxonomy of CP methods for text data. Loss bound refers to a formal guarantee on the maximum expected loss of a prediction set generated by a conformal method.

engender new challenges in CP due to their complex nature and unbounded output space. These LLMs are often trained on vast, inaccessible datasets and parameters, making re-training or even fine-tuning nearly impossible. CP addresses uncertainty without oversimplifying the complexity of LLMs and avoids the need for retraining and can be applied to any LLM. Primarily used in classification and regression tasks, harnessing the full potential of CP for text data and generative models such as LLMs is not trivial. In this section, we will explore CP in unstructured text data. We illustrate the structure of the CP method applied to NLP tasks in Figure 4. For a more comprehensive review of CP for NLP, please refer to the survey [26].

4.1.1 Text Classification. Text classification is often applied in high-stakes areas such as medical diagnosis, security, and legal document review. For example, Maltoudoglou et al. [124] build CP [140] upon the transformer method for sentiment analysis to address the problem of heavy computation of CP. Another work [70] similarly applied inductive Venn—ABERS predictor to transformers and constructed prediction sets based on conformal p -values calculated from estimated probabilities. Recent work focuses on the conditional coverage of CP for text classification, which focuses on the coverage for certain subpopulations (e.g., conditional on labels) and is particularly essential for unbalanced datasets and respective tasks. Giovannotti and Gammerman [72] compare several transformer-based conformal predictors and found that Mondrian CP demonstrated the smallest theoretically expected efficiency drop for conditional coverage of CP.

Another thread of work focuses on multi-label text classification, where each instance may have multiple labels, requiring CP to output a combination of labels. For example, [136] applies the power-set approach [137] to multi-label classification, which treats each label set as a candidate classification and is assigned a p -value. They also propose to only use the maximum observed label cardinality of the corpus to reduce the combination of labels. In the study [123], they extend the **Label Powerset (LP)** Inductive (i.e., Split) CP by reducing computational complexity, specifically by eliminating a significant number of label sets that are likely to have p -values below the specified miscoverage rate. [59] proposes a modified CP algorithm that produces multi-label prediction sets with a strict limit on false positives. By using a greedy algorithm based on the non-conformity scores, it optimizes true positive rates while controlling false positives below a user-specified level.

4.1.2 CP for Natural Language Understanding (NLU). NLU focuses on enabling machines to comprehend and interpret the structure and semantics of human language, it is becoming increasingly pivotal in various commercial settings. NLU involves analyzing text to extract meaning, intent, and context, handling ambiguity, and recognizing entities or relationships.

One application of NLU is QA tasks, which can be categorized into **Multiple Choice Question Answering (MCQA)** and open-domain QA. For the MCQA task, a common approach is to simplify it into a classification task for the given choices. Current approaches to applying CP to this task primarily rely on the standard split CP and developing various non-conformity scores for different needs. For example, one line of research [99, 201] involves appending the input with “The correct answer is the option:”, prompting the LLMs to predict the next word to acquire the logits of the option. Subsequently, a softmax operation is applied to these logits and then used for CP algorithm. In open-domain QA, an LLM receives a question and is asked to generate an answer from a vast, unbounded space based on its comprehension of the question. In a black-box setting, when sampled answers as the LLM output are available, CP can be used to construct a reliable set of potential answers. For example, Su et al. [170] employs sampling to assess the model’s uncertainty and calculates non-conformity scores based on frequency to construct prediction sets. They also distinguish the results using normalized entropy and semantic similarity as an extension. Another application of NLU in open-domain QA is to detect LLM hallucinations and verify the facts. Li et al. [109] construct a prediction set by sequentially building the retriever and response set with a coverage guarantee and then combining them into an aggregated set. Then it employs Bayesian optimization to enhance efficiency by finding the best hyper-parameters. An earlier work [60] first expands the CP correctness criterion to allow for additional, inferred “admissible” answers, which can substantially reduce the size of the predicted set while still providing valid performance guarantees. It then amortizes costs by conformalizing prediction cascades, in which it aggressively prunes implausible labels early on by using progressively stronger classifiers—again, while still providing valid performance guarantees.

Another three common NLU tasks where CP has been studied are *machine translation evaluation*, text infilling, and *part-of-speech (POS) tagging*. Machine translation evaluation involves assessing the quality of generations in a target language based on the input provided in a source language. Some studies apply the CP algorithm to machine translation evaluation to address uncertainty [71, 208]. For example, Zerva and Martins [208] show that split CP adjusts confidence intervals to achieve the desired coverage and identify biases in these intervals based on translation language pairs and translation quality. Using conditional CP, they create calibration subsets for each data subgroup, ensuring equalized coverage. In the POS tagging and text infilling, Dey et al. [42] propose CP-enhanced BERT and BiLSTM algorithms to ensure their statistical reliability. The non-conformity scores are designed based on multi-label classification tasks.

4.1.3 CP for Natural Language Generation (NLG). NLG is the process of transforming structured information or text into natural language output. It focuses on producing coherent, contextually appropriate language, such as generating summaries or translations.

In the field of NLP, transformer-based generative LLMs play a significant role [183]. These models generate text by predicting the next word based on previous inputs. When applying CP to text generation tasks, these tasks are categorized by the length of the generated content, distinguishing between next-word prediction and long sentence generation tasks. We note that different from MCQA, in the next-word prediction task, the label set encompasses the entire vocabulary, from which CP generates a prediction set containing potential next words.

To assess whether a top- p sampling set is indeed aligned with its probabilistic meaning in various linguistic contexts, Ravfogel et al. [150] leverage CP and customize the **Adaptive Prediction Sets (APS)** score [153], which aims to provide prediction sets that adaptively include all possible labels that meet a certain confidence level, ensuring that the true label is contained within the prediction set with a high probability. Another challenge in next-word prediction is that the complex intrinsic relationships within the data may violate the exchangeability assumption, leading to non-i.i.d. test and calibration data. To address this, Dennis Ulmer et al. [182] combine APS scores with non-exchangeable CP for next-word prediction.

Another increasingly popular NLG task is *code generation*, where LLMs are tasked with producing a piece of code based on input. One challenge lies in structured prediction problems, where the space of labels is exponential in size. To address this issue, Adam Khakhar et al. [97] propose generating code based on the abstract syntax tree structure and partial programs. They utilize partial programs for prediction [165] and establish a pre-selected 1D search space over the prediction sets by mapping features to labels in order to construct **Probably Approximately Correct (PAC)** prediction sets. **Retrieval Augmented Generation (RAG)** combines retrieval-based methods with LLMs for text generation. Integrating CP in its unique generation process is challenging. Kang et al. [96] propose to apply conformal risk control with RAG, in which they show that using RAG can lead to lower risk than normal LLM with or without covariate shift. Mohri et al. [131] address the problem of factual correctness in LLM outputs. During calibration, each output is broken into subclaims, scored by GPT-4 for confidence, and filtered using a threshold based on a preset confidence rate α . In the prediction stage, subclaims with confidence scores above this threshold are seen as correct factual and concatenated to form the final output.

Another line of research adopts CP to control the quality of LLM generations. To deploy the outputs of foundation models reliably, Gui et al. [78] present conformal alignment, which aims to provide guarantees that a prescribed fraction of selected model outputs are trustworthy and meet the alignment criterion. This is achieved by training an alignment predictor on reference data and selecting outputs whose predicted alignment scores exceed a calibrated threshold, thus controlling the FDR of untrustworthy outputs. Victor et al. [147] introduce a conformal sampling framework

for LLMs to generate diagnoses based on input content. Its goal is to generate prediction sets with provable guarantees of containing at least one acceptable response, while filtering out low-quality or redundant outputs. This is achieved through a calibrated sampling process that stops when confidence criteria are met, using rejection rules to eliminate undesirable samples, ensuring efficient, and reliable output.

4.1.4 Inference-time Efficiency. Besides generating prediction sets, CP can also improve inference-time efficiency by reducing the need for multiple samples or complex generation and leveraging efficient calibration strategies. Some works focus on achieving early stopping using CP, enabling the model to produce reliable predictions with reduced inference time. In [102], the authors propose training an LLM using minimal computational resources while maintaining accuracy. By leveraging the **Learn then Test (LTT)** framework, they identify optimal risk-controlling hyperparameter configurations. Their key contribution, Pareto Testing, further improves efficiency by strategically guiding the number and order of tested configurations when searching for valid settings. Schuster et al. [159] propose a novel method called **Confident Adaptive Language Modeling (CALM)**, which allows the transformer to exit early at a specific layer of the decoder, thereby significantly reducing inference time. This method employs various confidence measures to determine when to exit early, such as softmax response, hidden-state saturation, and a trained early exit classifier, then they use a calibration dataset to threshold confidence scores by grid search.

Discussion Although CP has recently achieved success in NLP, several questions remain to be addressed in the future: (1) Computational efficiency. The use of CP can be computationally expensive, especially with large datasets or packages like FAISS [93]. Optimizing CP for efficiency is essential. (2) Human and LLMs Interaction. As human-LLM interactions generate continuous new data, further research is needed to adapt CP to effectively and efficiently handle these unbounded output space and dynamic environments. (3) Ethical Concerns. CP's ability to produce multiple predictions increases the risk of generating unethical outputs, particularly in LLMs, and addressing these concerns remains an open challenge.

4.2 Image Data

Deep learning with images can achieve high predictive accuracy, but quantifying uncertainty remains a significant challenge, hindering its deployment in critical settings. Applying CP for image data involves navigating challenges related to high dimensionality and feature complexity. The following sections will introduce the development of CP in various **computer vision (CV)** tasks.

4.2.1 Image Classification. Some applications of CP for image classification are built upon the framework of canonical classification tasks, but further exploit the specific characteristics of image data across various domains to design powerful non-conformity score functions. For example, in [13], they aim to ensure the reliability of AI models in diverse clinical environments by introducing the conformal triage algorithm, which outputs labels {true, false, and unknown} to aid clinical decision-making. Another line of research utilizes learned deep (underlying) features of image data in constructing more efficient prediction sets. Teng et al. [179] extend the application of CP to semantic feature spaces, in which they leverage the difference between pre-trained and fine-tuned features to generate the prediction set via band estimation [197]. Ghosh et al. [66] further show that if learned image representation satisfies some mild conditions, their proposed NCP (neighborhood CP) approach will produce smaller prediction sets than traditional CP algorithms. They utilize the learned representation to identify the K -nearest neighbors of a given test input and assign important weights based on their distances to create adaptive prediction sets.

4.2.2 Image Segmentation. Given an image x as the input, image segmentation aims to find a map $f : x \mapsto y$, where y is the segmentation result as the output. Here x and y have the same dimension and for each pixel x_i , the corresponding value y_i represents the class for this pixel. As the softmax function has been shown to produce overconfident predictions with uncalibrated outputs, CP can be beneficial in calibrating the segmentation approaches. [191] combines deep learning with CP for tissue sub-region prediction with confidence. They define the non-conformity score based on the output softmax score for each pixel. They further use the conformal p -values to guide prediction: they first choose the class with the largest p -value as the predicted output. Then, confidence in the prediction is defined as the gap between the largest and the second largest p -values, with which the pixel with confidence lower than a given confidence level is excluded. In addition, a recent work [132] utilizes the framework of conformal risk control, where the nested sets $\{\widehat{C}_\lambda(\cdot)\}_{\lambda \in \Lambda}$ is defined by thresholding the output softmax scores by λ . The scarcity of images for pixel-level calibration makes it hard to obtain sufficient images for pixel-level calibration. Brunekreef et al. [24] introduce the concept of Kandinsky calibration, which leverages the inherent spatial structure within natural image distributions, to cluster pixels.

4.2.3 Image-to-image Tasks. In image-to-image tasks, including inpainting, super-resolution, and colorization, the objective of CP is to construct a prediction interval around each pixel such that the true pixel value falls within this interval with a user-defined probability.

Canonical approaches to this task define uncertainty regions based on probable values per pixel, while ignoring spatial correlations within the image, resulting in an exaggerated volume of uncertainty. To address this, one study [157] quantifies the uncertainty in a latent space, thus taking spatial dependencies into account. Specifically, they aim to develop a method that can provide principled uncertainty intervals for semantic factors within a disentangled latent space of a generative model. This method has a fixed decoder that can predict images based on latent code with conformal quantile regression and uses a decoder to generate restored images from the prediction set. However, by relying on a non-linear, non-invertible and uncertainty-oblivious transformation, this method suffers from interpretability limitations. Belhasin et al. [20] improve this by using spatial relationships between pixels for UQ in inverse imaging problems. They employ a diffusion model with three phases: (1) approximation, where sampling generates principal components and weights using SVD; (2) calibration, where three variants of **Principal UQ (PUQ)** are proposed: exact PUQ, Dimension-Adaptive PUQ, and Reduced Dimension-Adaptive PUQ, with the latter two improving computational efficiency; and (3) using the previous result for prediction.

There are several works that focus on conformal risk control for image-to-image tasks. For example, [11] proposes a distribution-free pipeline for image-to-image regression based on risk control [18]. [101] proposes a method for conformal risk control based on masking, which guarantee the masked reconstructed image and masked true image have a distance under threshold in a given probability. [168] proposes a novel method for applying diffusion models to CP. Their approach involves generating multiple reconstructed variations for each original image and then determining the interval bounds by taking the upper and lower quantiles for each pixel across these variations. To deal with the problem that classical risk control can only be applied to scalar hyper-parameters, in the study [178], they propose entrywise calibrated intervals and multi-dimensional risk control.

Discussion CP for image classification focuses on high efficiency, handling data imbalance or noise, and achieving the performance of full CP. These methods perform well because they address various challenges associated with CP. In image-to-image tasks, encoder-decoder structures like diffusion models are commonly used. However, there are still some problems that can be better

addressed. For example, in image-to-image tasks, leveraging the relationship between the prediction sets of different pixels to improve predictions is an area for further exploration.

4.3 Heterogeneous Data

Heterogeneous data consists of diverse types and formats, often originating from different sources. This diversity presents challenges for traditional CP methods, which are generally designed for more uniform data types. The variability in distributions, complexity in feature interactions, and the need to handle different data structures simultaneously make it difficult for traditional CP to provide accurate and reliable prediction intervals. Therefore, new CP methods are required to effectively manage heterogeneous data, ensuring that predictions remain reliable and maintain confidence guarantees across all types of data involved. For heterogeneous data, the objective is to find a function that can map $\mathcal{X} \rightarrow \mathcal{Y}$, where $\mathcal{X} = \mathcal{X}_1 \cup \dots \cup \mathcal{X}_K$. Then in a standard CP method for heterogeneous data, for each sample x , a prediction interval/set is generated by:

$$C(x) = [\hat{l}, \hat{u}] \text{ or } [c_1, c_2, \dots, c_n], \quad (32)$$

where $\hat{l}$ and $\hat{u}$ are the lower and upper bounds of the predicted interval, respectively, c_i is a label.

One study [181] works on multi-modal data and highlights that previous methods for generating multi-modal prediction regions often result in overly complex solutions. To address this, it proposes using shape templates—such as ellipsoids, convex hulls, and hyperrectangles—over the residuals of calibration data, which simplify the non-conformity score functions, leading to more efficient and accurate prediction regions. When there is noise in the multi-modal data, Zhan et al. [209] propose a reliability-based training data cleaning method using split CP, which leverages well-labeled data as a calibration set to detect and correct mislabeled data and outliers. In another study [50] for image caption, they introduce a novel heuristic for zero-shot CP that leverages CLIP-based foundation models [148] calibrated by web data. Some work develops CP for limited data scenarios with multi-modal training, e.g., zero-shot, one-shot, and few-shot learning. For example, Fisch et al. [58] point out that when the dataset is small, the prediction generated by CP will be unusually large. They address it by casting CP as a meta-learning paradigm over exchangeable collections of auxiliary tasks. Another type of heterogeneous data is multi-view data which refers to datasets where the same underlying object or phenomenon is described by multiple different sets of modes or “views.” **Multi-View CP (MVCP)** [151] addresses the challenge that the naive method for late fusion fails to leverage the uncertainty information of individual predictors, leading to less efficient and adaptive prediction intervals. Specifically, they define the score function for the predictor in each mode and use quantiles along projection directions. The final prediction region is an intersection of half-planes, ensuring efficient, distribution-free coverage.

Another application of heterogeneous data is in federated or transfer learning, where the model processes multi-source inputs. In [121], the authors highlight that data in federated learning is typically non-IID, as data from different sources often follow distinct distributions. This characteristic poses challenges to the non-exchangeability assumption of CP. To address this issue, they propose the concept of an ϵ -approximate β -quantile, which allows the quantile to deviate slightly from the target value while maintaining similar coverage properties. Within this framework, non-conformity scores are computed locally for calibration before communication occurs. Further, in transfer learning, Liu et al. [116] study covariate shifts when data originates from multiple sources. The proposed data-adaptive strategy begins by calculating site-specific quantiles for each source. For the target distribution, it then computes a discrepancy measure between the target site’s quantile and each source site’s quantile. This discrepancy is used to assign weights and minimize the discrepancy for each source site and the target site.

Table 1. A Summary of Representative CP Methods for Spatio-temporal Data Described in Section 5

Representative Work	Data	Distribution Assumption	Coverage Guarantee
NexCP [16]	None	None	$\geq 1 - \alpha - \sum_{i=1}^n \tilde{w}_i \cdot d_{TV}(R(Z), R(Z^i))$
EnbPI [194]	single time series	strongly mixing errors	$\approx 1 - \alpha$
SPCI [196]	single time series	None	$\approx 1 - \alpha$
ACI [68]	single time series	None (online)	Asymptotically $\approx 1 - \alpha$
CF-RNN [169]	independent time series	exchangeable time series	$1 - \alpha$
Copula-RNN [174]	independent time series	exchangeable time series	$1 - \alpha$
MultiDimSPCI [192]	dependent time series	non-exchangeable time series	$\approx 1 - \alpha$

Discussion Existing CP methods for heterogeneous data are relatively limited, primarily focusing on multi-modal and multi-view data, as well as scenarios like covariate shift and limited data availability. Future research could explore leveraging contrastive learning frameworks, similar to CLIP for multi-modal image captioning, to enhance conditional probability CP efficiency. Another promising direction is addressing covariate shifts across different modalities in multi-modal tasks, which remains largely unexplored. Further, managing uncertainty in multi-modal outputs, which often arises from the complexity of such data, presents a compelling challenge that warrants further investigation. Finally, more efforts are needed to explore CP in the semantic space.

5 Dynamic Data—Spatio-temporal Data

Spatio-temporal data captures phenomena that evolve over time and across locations, making it crucial in fields like meteorology, geosciences, urban planning, and public health. As prediction models for spatio-temporal data are often setting-specific, prediction sets instead of point predictions can be preferred in this scenario. Applying CP to this type of data helps manage risks associated with spatial and temporal variability by providing confidence measures for predictions. However, standard CP assumes data exchangeability, an assumption often violated by spatio-temporal data due to its structured dependencies and evolving nature. This can lead to misleading confidence measures if not properly addressed. This section discusses challenges and methodologies associated with applying CP to spatio-temporal data. We begin by discussing one-dimensional spatio-temporal data, which essentially constitutes time series data collected from a single geographic location. Then, we explore strategies for tackling multi-location (i.e., multi-dimensional) spatio-temporal data, where each entire series is treated as a single observation, exhibiting temporal dependencies both within and between the sequences. Therefore, the violation of exchangeability in multi-dimensional settings can be more severe and requires robust analytical approaches to ensure the accuracy and reliability of predictive models. In addition, we will cover the special case of streaming time series data, highlighting the need for new methods and identifying additional areas for the application of CP. We summarize representative CP methods in this section in Table 1 and Figure 5.

5.1 One-Dimensional Spatio-temporal Data

For one-dimensional spatio-temporal data, there are three primary lines of research:

- **Reweighting** adjusts the influence of historical samples based on their age or relevance, addressing the non-stationarity. By altering the weights assigned to each sample, sample reweighting helps to align the distribution of the weighted calibration data more closely with that of the test data, ultimately addressing the issue of distribution shift.
- **Updating non-conformity scores** updates non-conformity scores continuously as new data becomes available, ensuring that the model adapts to recent trends and patterns without being overly influenced by outdated information.

- **Updating α :** Instead of using a fixed miscoverage rate for all predictions, this method adjusts the miscoverage rate based on recent prediction errors and variability in the data. Such adaptivity allows for more flexible and context-sensitive prediction intervals, crucial for applications where sudden changes in volatility or trend are common.

5.1.1 Reweighting. Reweighting is one of the most common frameworks for non-exchangeable settings, including time-series data. As we show in Section 2.3, weighted CP can compute prediction intervals for problems with covariate shifts between training and test data [180]. Below, we discuss several reweighting approaches that can be applied to one-dimensional time series data.

Pre-defined weights To relax the covariate shift assumption and address the problem that the high-dimensional likelihood ratio is known beforehand or well approximated for correct coverage in weighted CP, Barber et al. [16] extended the weighted method in [180] to the scenario with arbitrary pre-defined weights, showing that the coverage gap can be bounded by the total variation between the calibration and test data distributions. The weights $\{w_1, \dots, w_t \in [0, 1]\}$ for non-conformity scores are normalized as follows:

$$\tilde{w}_i = \frac{w_i}{w_1 + \dots + w_t + 1}, \quad i = 1, \dots, t, \quad \text{and} \quad \tilde{w}_{t+1} = \frac{1}{w_1 + \dots + w_t + 1}. \quad (33)$$

The weighted non-conformity scores for split and full CP are then:

$$\sum_{i=1}^t \tilde{w}_i \cdot \delta_{V_i} + \tilde{w}_{t+1} \cdot \delta_{\infty} \quad \text{and} \quad \sum_{i=1}^{t+1} \tilde{w}_i \cdot \delta_{V_i}^y, \quad \text{respectively.} \quad (34)$$

Here, $V_i = |Y_i - \hat{f}(X_i)|$ is the non-conformity score for the trained model $\hat{f}$. Let d_{TV} denote the total variation distance between distributions, theoretically, [16] provided the lower bounds for coverage in non-exchangeable CP:

$$\mathbb{P} \left\{ Y_{n+1} \in \hat{C}_n(X_{n+1}) \right\} \geq 1 - \alpha - \sum_{i=1}^n \tilde{w}_i \cdot d_{TV} (R(Z), R(Z^i)). \quad (35)$$

This approach can be applied to non-symmetric algorithms like weighted regression and autoregressive models, which are common in time series forecasting. Building upon [16], subsequent research broadened the framework's applications. For example, Schlembach et al. [158] expanded the methodology for multi-step-ahead time series forecasting to the multivariate setting and conformal risk control measures were incorporated into one-dimensional spatio-temporal data [55].

Learned weights The seminal work [16] uses pre-defined weights, leading to poor generalization. HopCPT [14] addresses this by using a **Modern Hopfield Network (MHN)** for similarity-based reweighting, which enhances accuracy by prioritizing past instances that closely resemble current conditions [14, 149]. HopCPT assigns weights in time series forecasting, where each weight $a_{t,i}$ signifies the relevance of a past time step i to the current time step t . The weights and corresponding normalized weights are computed as follows:

$$a_{t,i} = \beta m(z_t)^\top W_q W_k m(z_i), \quad \tilde{a}_{t,i} = \frac{e^{a_{t,i}}}{\sum_{j \in M} e^{a_{t,j}}}, \quad \forall i \in M, \quad (36)$$

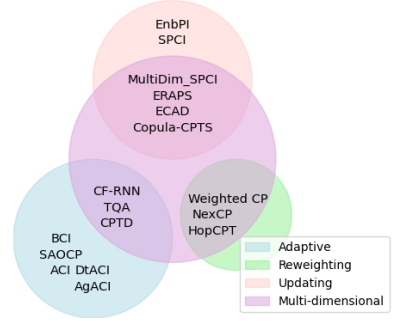


Fig. 5. Venn diagram for CP methods in spatio-temporal data.

where z_i is the encoded input for time step i , m is an encoding function, and W_q, W_k are matrices of learned weights. The hyperparameter β adjusts the focus of the probabilistic distribution generated by the softmax function. Using these weights, HopCPT generates the prediction interval by discounting the quantiles of relative errors:

$$\widehat{C}_\alpha(x_t, M) = \left[\widehat{f}(X_t) + Q_{\frac{\alpha}{2}} \left(\sum_{i \in M} \widetilde{a}_{t,i} \delta_{\epsilon_i} \right), \widehat{f}(X_t) + Q_{1-\frac{\alpha}{2}} \left(\sum_{i \in M} \widetilde{a}_{t,i} \delta_{\epsilon_i} \right) \right], \quad (37)$$

where Q_τ denotes the τ -quantile and δ_{ϵ_i} is a point mass distribution of the error ϵ_i at time step i . To train the MHN, an auxiliary task using absolute errors as proxy values is employed, aligning errors from comparable regime time steps to predict the current absolute error.

Discussion HopCPT exemplifies cutting-edge CP techniques for time series data, offering remarkable forecasting precision [14]. Future research can focus on refining weight selection mechanisms to better capture temporal dependencies in data (e.g., [31]). Advancing this aspect of CP can lead to significant theoretical innovations and enhance practical forecasting performance.

5.1.2 Updating Non-conformity Scores. In contrast to reweighting methods that adjust weights based on a predefined strategy, this section introduces updating non-conformity scores to address non-stationary time-series data. By leveraging the most recent T samples and continuously updating prediction intervals as new data becomes available, this approach adapts the intervals to reflect the latest data trends. For example, when data is continuously streamed, updating non-conformity scores uses only the latest data points to calculate prediction intervals, quickly adapting to new trends or shifts in the data. Therefore, updating non-conformity scores dynamically integrates new information without segregating data into calibration and test subsets.

A pioneering work proposes EnbPI [194], which uniquely updates the non-conformity score with sequential error patterns to adapt the intervals dynamically. This approach starts with bootstrap training data to create effective “**leave-one-out**” (LOO) ensemble prediction models with input $\widehat{X}$: $f_{\widehat{X}}(\cdot) = \phi(\{f_b(\cdot) : t \notin S_b\})$, where ϕ denotes an arbitrary aggregation function (e.g., mean, median) over a set of scalars, and $S_b \subset [T]$ is the bootstrap index set used to train the b -th bootstrap estimator f_b . The point predictor on test data is defined as $f_{\widehat{X}}(X_t)$, which aggregates all bootstrap predictions, reducing the possibility of overfitting. EnbPI then updates the past residuals during predictions to ensure the prediction intervals have adaptive width. For a fixed $h \geq 1$, define $\epsilon_t^h := \{\widehat{\epsilon}_1, \dots, \widehat{\epsilon}_{t-h}\}$, where the residuals are $\widehat{\epsilon}_t = Y_t - f_{\widehat{X}}(X_t)$. The prediction intervals $C_{t-1}(\widehat{X})$ are given by:

$$\left[f_{\widehat{X}}(X_t) + Q_{\alpha/2}(\epsilon_t^T), f_{\widehat{X}}(X_t) + Q_{1-\alpha/2}(\epsilon_t^T) \right], \quad (38)$$

which utilize the past $h = T$ residuals and resemble traditional CP intervals due to the use of the empirical quantile function $Q_{1-\alpha/2}$ to compute interval width. This method offers computational efficiency, robustness against overfitting, and adaptability by avoiding the need to retrain models, instead using LOO ensemble models [194].

The approximately valid coverage guarantee of the EnbPI method is contingent on the assumption that the error terms $\{\epsilon_t\}_{t \geq 1}$ are both stationary and strongly mixing. However, this assumption may not hold in time-series models. SPCI [198] advances beyond EnbPI by relaxing the strict assumptions on error distributions. SPCI incorporates quantile regression to effectively address serial dependencies among residuals in sequential analysis. It replaces the empirical quantile with an estimate by a conditional quantile estimator and adjusts the prediction interval dynamically by optimizing its width. This is done by selecting a threshold parameter that minimizes the gap between two conditional quantile estimates while ensuring the desired coverage level.

Follow-up studies on SPCI have introduced more sophisticated methods to handle the dependent structure of the time-series data. For example, Lee et al. [103] applies a reweighted Nadaraya-Watson estimator within the realm of nonparametric kernel regression, thus enhancing the selection of data-dependent optimal weights. Additionally, Lee et al. [104] adopts a Transformer decoder as a conditional quantile estimator in the SPCI framework, recognizing its effectiveness in sequential modeling tasks. Both methods facilitate the computation of weights in a data-driven manner, eliminating the need for prior knowledge about the data.

Discussion The “updating” approach for time series is akin to reweighting, but it uniquely utilizes data-adaptive weights. By incorporating the most recent T samples into the calibration set, this method ensures that the weights adjust dynamically to reflect the latest data trends and patterns. This similarity to reweighting, yet with a focus on adaptive recalibration, maintains the model’s relevance and accuracy amidst evolving data. Future enhancements could involve developing more sophisticated algorithms to determine optimal T values are based on real-time changes in data volatility or distribution patterns. Integrating predictive analytics to anticipate the need for updates could further refine its effectiveness.

5.1.3 Updating the Miscalibration Rate. Another direction for CP in time series focuses on adaptively adjusting the miscalibration rate α during test time to account for mis-coverage. This method can dynamically adjust the size of prediction sets in an online setting where the data-generating distribution varies over time in an unknown fashion. Such an approach is particularly advantageous for time series data. The continuous recalibration during the testing phase ensures that the prediction intervals maintain an appropriate level of coverage over time.

ACI [68] is designed for constructing prediction intervals for time series data in an online learning framework. The approach is simple in that it requires only the tracking of a single parameter that models the shift. The update rule for the significance level α is: $\alpha_{t+1} = \alpha_t + \gamma(\alpha - \text{err}_t)$, where γ is a step size parameter, and err_t indicates if Y_t is excluded from $\hat{C}_t(\alpha_t)$. The approach ensures that the prediction intervals adjust over time to account for shifts in the data distribution, maintaining the desired coverage probability.

The choice of γ gives a tradeoff between adaptability and stability. While raising the value of γ will make the method more adaptive to distribution shifts, it will induce greater volatility in α_t . Zaffran et al. [204] theoretically analyzed the influence of γ on ACI efficiency and proposes AgACI which runs multiple ACI instances with different learning rates and combines their outputs using an online aggregation of experts approach. Alternatively, Gibbs and Candès [69] dynamically adjusts γ based on expert performance. Each expert operates with a specific γ , and ACI is performed in parallel for all experts. The final α_t is selected based on the historical performance of each expert, effectively learning the optimal γ over time to adapt to shifts in the data distribution.

Another alternative to ACI is the **multi-valid CP (MVP)** method of [17] which accounts for worst-case adversarial guarantees to ensure the methods remain reliable even when data distributions are manipulated or shift unexpectedly, especially in adversarial environments. MVP achieves this by selecting prediction thresholds that control the long-term miscalibration rate. Similarly, Angelopoulos et al. [7] address worst-case scenarios by examining the effect of a decaying step size on the prediction threshold update rule. Their approach ensures strong long-term coverage in adversarial settings and achieves convergence to an optimal threshold.

Another follow-up work of ACI focuses on refining the regret condition within the online learning framework. SAOCP [17] targets minimizing **Strongly Adaptive Regret (SAReg)**, which evaluates performance over all intervals of specific lengths rather than over the entire length. In [6], control theory is used to dynamically manage uncertainties and adapt to changing data. It integrates ACI with traditional PID control mechanisms to enhance predictions. Additionally, Yang et al. [200]

proposes BCI to address ACI's limitations by incorporating **Model Predictive Control (MPC)** techniques to optimize prediction intervals. Together, these advancements extend ACI's capabilities by enhancing adaptability and robustness in dynamic and adversarial environments.

Discussion Approaches that dynamically adjust the miscoverage rate based on incoming data, akin to how online learning updates model parameters in response to new information. Future research could explore more nuanced methods for adjusting miscoverage rate, potentially incorporating reinforcement learning to optimize decision-making processes based on performance feedback. Importantly, these methods can be synergistically combined; for instance, updating may be viewed as a specialized form of reweighting that places greater emphasis on recent samples. Moreover, the integration of reweighting with adaptive strategies has been shown to enhance performance significantly [14, 79]. While the strategies outlined are broadly applicable to spatio-temporal data, research exists for specific types of time series, such as AR process [204], Hidden Markov Model [133], and β -mixing process [134], which further refine these general techniques.

5.1.4 Other Methods. Chernozhukov et al. [34] develops a randomization method and accounts for potential serial dependence by including block structures in the permutation scheme, which can be either overlapping or non-overlapping. The non-overlapping blocking scheme divides the data into separate blocks to reduce dependencies within each block. In contrast, the overlapping blocking scheme shifts the starting point of each block slightly to create overlapping sections between them. This overlap allows the scheme to capture more variations and dependencies between observations by including samples that are close together in multiple blocks. Follow-up work such as [2] employs similar blocking techniques to account for dependencies in a probabilistic forecasting scheme for two-dimensional functional time series. Cleaveland et al. [37] propose an optimization-based method that reduces the conservatism of CP for time series predictions. Instead of treating prediction errors independently at each time step, they consider a parameterized prediction error over multiple time steps and optimize the parameters over an additional dataset. Prinzhorn et al. [145] further analyzes different temporal components using time series decomposition. This method decomposes the original time series into remainder, trend, and seasonal components and leverage previous methods such as EnbPI [194] and ACI [68] to handle these different components. This strategy better captures the inherent characteristics of time series and improves performance.

A very recent work by [135] extends split CP to handle non-exchangeable data by introducing a novel framework based on concentration inequalities and data decoupling properties. The method assumes that the calibration and test data distributions are reasonably similar and stable, ensuring reliable prediction intervals with guarantees for both marginal coverage and conditional coverage.

5.2 Multi-Dimensional

Multi-dimensional spatio-temporal data spans multiple dimensions of both time and space, introducing additional complexity and opportunities for predictive modeling. The dependencies between spatial and temporal dimensions further undermine the exchangeability assumption inherent in traditional methods. Previous works on multi-dimensional data primarily focus on exchangeable data [56, 67, 94, 130], overlooking the temporal and spatial dependencies. This section mainly focuses on advanced techniques that adapt CP for multi-dimensional spatio-temporal data in regression problems.

A simple approach is to treat each time step as a separate prediction task. CF-RNN [169] constructs prediction intervals using Bonferroni correction, selecting the $(1 - \alpha/k)$ th quantile for a confidence level of $1 - \alpha$. This method results in conservative intervals and ignores spatial dependencies. Similarly, Mao et al. [125] achieves theoretical validity by assuming that spatial locations are exchangeable. Without making this assumption, they focus on specific locations and propose

a localized approach that utilizes nearby locations to address spatial dependence. Zhou et al. [212] also assume that each time series trajectory is exchangeable, combining with ACI to produce more informative prediction intervals for motion planning. Sun and Yu [174] considers spatial dependence by using copulas to model uncertainty over future time steps, reducing confidence interval widths while maintaining validity. Copulas, which model dependencies between variables, apply the empirical **cumulative distribution function (CDF)** to estimate non-conformity scores, ensuring the predicted trajectory falls within the confidence intervals at a level of $1 - \alpha$.

The theory behind [174] assumes that each data sample in a time series is i.i.d., thus ignoring temporal dependencies. Xu et al. [192] relax this assumption by constructing episodic prediction sets that are adaptively and efficiently calibrated during test time to account for temporal dependencies. This method starts with a training dataset $Z_t = (X_t, Y_t)$ for $t = 1, \dots, T$, and a prediction algorithm $\hat{f}$ that forecasts Y . Key to this method is the computation of prediction residuals $\hat{\sigma}_t = Y_t - \hat{f}(X_t)$, which are used to estimate the covariance $\hat{\Sigma}$, because it captures the dependencies and relationships between different dimensions of the data. The covariance is:

$$\hat{\Sigma} = \frac{1}{T-1} \sum_{t=1}^T (\hat{\sigma}_t - \bar{\sigma})(\hat{\sigma}_t - \bar{\sigma})^T, \quad (39)$$

where $\bar{\sigma}$ is the mean of $\hat{\sigma}_t$. They ensure the invertibility of $\hat{\Sigma}$ via a low-rank approximation, crucial for defining the ellipsoidal (prediction) region $B(r, \bar{\sigma}, \hat{\Sigma}_\rho)$ with a radius r . This radius is calibrated through CP by calculating a non-conformity score for a new sample (X, Y) :

$$\hat{e}(Y) = (\hat{\sigma} - \bar{\sigma})^T \hat{\Sigma}_\rho^{-1} (\hat{\sigma} - \bar{\sigma}), \quad (40)$$

where $\hat{\sigma} = Y - \hat{f}(X)$. The method then determines r to ensure predictions fall within the ellipsoid at a specified confidence level. The prediction sets are then constructed via SPCI [192, 196].

5.2.1 Cross-sectional Coverage. In the multi-dimensional time series domain, an interesting branch considers both cross-sectional and longitudinal validity simultaneously. Two works [112, 113] enhance prediction intervals in time series forecasting by ensuring both cross-sectional and longitudinal coverage. Lin et al. [113] introduce **Temporal Quantile Adjustment (TQA)**, which dynamically adjusts quantiles using temporal information. TQA employs a two-step strategy: predicting future quantiles based on non-conformity scores and refining them through a budgeting strategy (TQA-B) or error-based adjustment (TQA-E). By contrast, Lin et al. [112] focus on normalization to account for distribution changes. It calculates these scores based on historical data discrepancies and normalizes them with the median absolute deviation to adjust for shifts in data distribution over time.

5.2.2 Classification and Anomaly Detection. Classification and anomaly detection within spatio-temporal data contexts also hold significant importance. Building upon their foundational research, Xu and Xie [195] developed the **Ensemble Regularized Adaptive Prediction Set (ERAPS)**, which adapts the EnbPI framework [194] for classification problems by applying the **Regularized Adaptive Prediction Set (RAPS)** method [5] from image classification to construct non-conformity scores. Additionally, [193] further developed EnbPI for anomaly detection in multi-dimensional data with ensembling to obviate the need for data splitting and enhance prediction accuracy and robustness.

Discussion This section discusses methods for handling dependencies in time and space, highlighting the challenge of solving non-exchangeability issues. Additionally, simultaneously considering longitudinal and cross-sectional coverage is an area that requires further exploration. Finally,

the use of graphical models for multi-dimensional time series remains underexplored, presenting a promising direction for future research.

5.3 Streaming Data

Streaming data is characterized by continuous flow, high volume, and rapid updates. **Adaptive CP (ACP)** methods [68, 69] for streaming data dynamically adjust α to maintain desired miscoverage rate as data characteristics evolve. In this section, we first introduce the application of ACP in motion planning. Next, we discuss the use of CP for detecting concept drift and enhancing interpretability in streaming data and finally extend CP to online learning beyond numerical data streams.

5.3.1 Streaming Numerical Data. Motion planning involves navigating agents from an origin to a destination without collisions, despite uncertainties from environmental factors, sensor errors, and model inaccuracies. The environment includes N dynamic agents with states Y_t following an unknown distribution $\mathcal{D}$. ACP manages streaming data to forecast future states over a prediction horizon H , updating the adaptive threshold δ_τ based on prediction accuracy:

$$\delta_{\tau+1} := \delta_\tau + \gamma(\delta - e_\tau), \quad (41)$$

where e_τ indicates prediction accuracy, and γ is a tuning parameter. Using ACP, an Uncertainty-Informed MPC optimizes a control sequence to ensure safe navigation by integrating real-time data and adapting to dynamic environments [48, 114].

CP can also help monitor online data streams, particularly in tasks like concept drift detection [53, 122, 177] and regression trees [90–92]. For instance, **Inductive CP (ICP)** assigns non-conformity scores to new samples based on a model trained on initial data. If the p-value for a new sample falls below a certain threshold, indicating a potential concept drift, a new classifier is trained; otherwise, the model is updated with the current labeled data. Additionally, CP enhances both interpretability and efficiency in regression trees by applying a Mondrian conformal predictor at each leaf, focusing on localized updates to maintain accuracy and adapt to changes in the data stream.

5.3.2 Beyond Numerical Data. Other studied problems with streaming data include image classification and document retrieval in online learning. Ge et al. [64] explore the use of stochastic online CP with semi-bandit feedback in real-time data streams, where the system only receives information about the true label if it is included in the prediction set. Stochastic online CP updates its prediction thresholds using the **Dvoretzky-Kiefer-Wolfowitz (DKW)** inequality to ensure the empirical distribution of non-conformity scores up to time t is not overly optimistic about the true distribution of the data by adding an adjustment factor. The adjusted threshold ensures that the prediction intervals remain conservative enough to maintain the desired coverage.

Discussion The seminal work ACI can be a general strategy to handle high-volume, non-stationary streaming data. When applying CP for streaming data, it is crucial to consider computational efficiency. Additionally, there is a significant need for more research on scalability, particularly in distributed systems where data streams are processed in parallel across multiple nodes. Developing efficient algorithms that can maintain statistical guarantees while operating in such environments would be highly beneficial. Finally, exploring the application of CP in multi-modal data streams, where data comes from various sources and types, presents another exciting research frontier.

6 Open Challenges and Future Directions

Non-exchangeability: distribution heterogeneity and temporal dependence. The exchangeability assumption often fails in real-world scenarios. While existing methods like non-exchangeable CP [16] assess the coverage gap of weighted CP based on the distance between distributions, developing an adaptive approach to minimize this gap remains an open challenge. Moreover, works such as ACI and related studies [7, 68] provide adversarial coverage guarantees for sequences with arbitrary distributions, but they do not ensure validity at each step. Given this, it would be valuable to explore how light distributional assumptions on the sequence could enhance adaptivity.

Robust CP with more general data imperfections. Ensuring the reliability of CP methods under data imperfections and adversarial conditions is crucial. As highlighted in the survey, critical scenarios include observations with missing values, noisy entries, or adversarial attacks. The validity of some existing methods hinges on specific distributional assumptions, such as the missing-at-random assumption. However, real-world missingness can follow more complex patterns, like block-wise or non-random missingness. Extending CP to handle such general data imperfections, while maintaining prediction set validity with tractable coverage gaps, remains an open problem.

Handling emerging data types. As data evolves, new data types bring challenges and opportunities for CP. For example, multi-modal data, which integrates information from various sources such as text, images, and sensor data, presents the challenge of combining heterogeneous data sources with differing noise characteristics and distributions. A key open question is how to “optimally” aggregate data from multiple modalities to enhance CP performance. Similarly, the rise of streaming data underscores the need for adaptive and online CP techniques that can provide valid predictions in real time. This requires maintaining computational efficiency and prediction validity as new data arrives and as the underlying data distribution potentially shifts over time.

CP for responsible AI. CP offers a promising approach to responsible AI [118]. CP can generate clear, quantifiable confidence measures, aiding users in understanding and trusting model predictions, which promotes interpretability [87, 90, 91]. For fairness, CP can help equalize coverage across diverse population groups, correcting biases without the need for expensive retraining [120, 153, 189]. Privacy is another critical area, especially in federated learning [119, 121]; CP can be adapted to produce confidence intervals without accessing sensitive or personally identifiable information, thus preserving privacy while ensuring reliable predictions. By addressing these aspects, CP can significantly contribute to the development of AI systems that are not only accurate but also fair, transparent, and resilient across various applications.

CP in aiding decision-making. Research indicates that when humans are provided with CP sets as *side information*, their task accuracy improves compared to using fixed-size prediction sets, thereby enhancing human-in-the-loop decision-making and human-AI collaboration [40]. CP also aids in early stopping for iterative algorithms by providing robust uncertainty measures, which helps prevent overfitting and reduces computational costs [111]. In online auctions, CP can predict final prices and dynamically adjust bidding strategies, leading to improved efficiency and profitability [80]. These applications highlight CP’s potential to transform various fields through reliable and adaptive predictions. Thus, it is essential to explore additional scenarios where CP can be applied to further unlock its potential and broaden its impact.

CP for cross-disciplinary research. Cross-disciplinary research offers significant opportunities to advance CP. For example, insights from statistics can contribute to developing new

theoretical foundations and methods for CP. Collaborations with domain experts in areas like healthcare, finance, and environmental science can lead to the creation of application-specific CP methods that address unique challenges in these fields. Additionally, leveraging advances in computer science, particularly in optimization and high-performance computing, can improve the scalability and efficiency of CP techniques. Interdisciplinary research that bridges theory and practice is also crucial for translating CP methods into real-world applications, thereby maximizing their impact.

Open source tools for CP. Despite progress in developing open-source tools for CP, challenges remain in expanding their usability. Current tools like PUNCC [128] and TorchCP [190] address traditional tasks such as regression, classification, and graph-based learning, while Fortuna [41] takes a broader approach, incorporating UQ techniques such as Bayesian inference. Additionally, Susmann et al. [175] introduced an open-source R package for ACI. While many tools effectively handle traditional data types, there is a notable lack of support for modern data types, such as unstructured data and streaming data, and for addressing the rapid growth of language modeling applications. Improving compatibility, scalability, and ease of use across ML platforms is crucial to foster wider adoption and innovation in CP.

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A Appendix

Table 2. Major Notations Used Throughout the Survey

Notation	Description
$X_i, \mathcal{X}$	feature of i th instance and feature domain
$Y_i, \mathcal{Y}$	label of i th instance and label domain
$V(X_i, Y_i), V_i$	non-conformity score of i th instance
$Q_{1-\alpha}, \hat{Q}_{1-\alpha}$	quantile or estimated quantile at $1 - \alpha$
α	miscoverage rate
$C_{1-\alpha}(X_i)$	predicted set of i th instance
D, n	dataset and its size
σ_i	residuals or estimated residual of i th instance
$P, P_X, P_{Y X}$	distribution, distribution of X , distribution of Y given X
$\hat{F}(X_i, Y_i)$	estimated CDF
π	permutation
θ_Z	parameter of model get from data Z
z_i, Z	pair of label and feature, equal to (X_i, Y_i) or (X, Y)
ℓ	loss function
$\hat{\tau}$	calibration threshold
t, T	time stamp
f, g	neural network or predictor
$\hat{l}, \hat{u}$	lower and upper bound of prediction interval
w_i	weight for i th instance
δ_{R_i}	point mass of i th instance

Received 28 September 2024; revised 17 February 2025; accepted 1 April 2025